

Big data analytic

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Dimensionality Reduction



Acknowledgement

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- The lecture uses a portion of slides from “Mining of massive datasets” copyrighted by Jure Leskovec, Anand Rajaraman, Jeff Ullman, Stanford University.
- <http://www.mmds.org>

Curse of dimensionality

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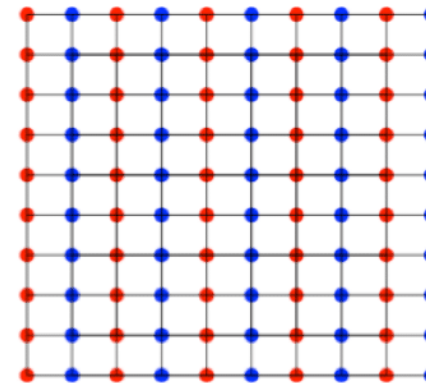
- **When data has too many features**
- **Disadvantages:**
 - Sparsity of data
 - Irrelevant data
 - Complex and costly model for data analytic
 - Difficulties in data clustering
 - Difficult to visualize
 - Difficult to find a solution
 - Overfitting the model

Dimensions grows, dimensions space increases exponentially

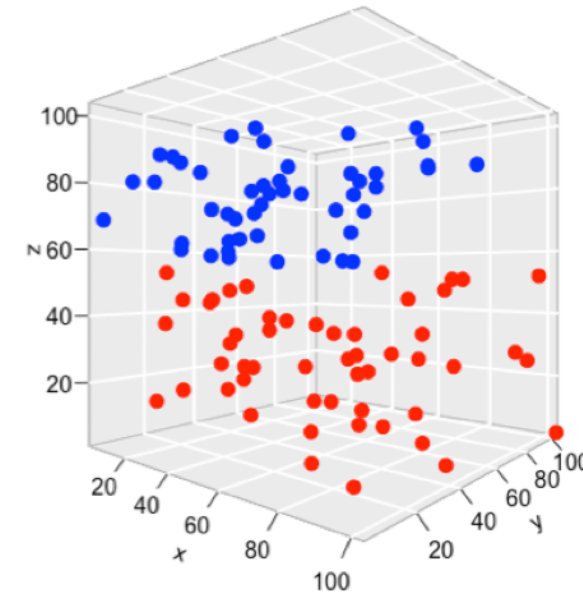
Source: Genomics



(A) 1-D



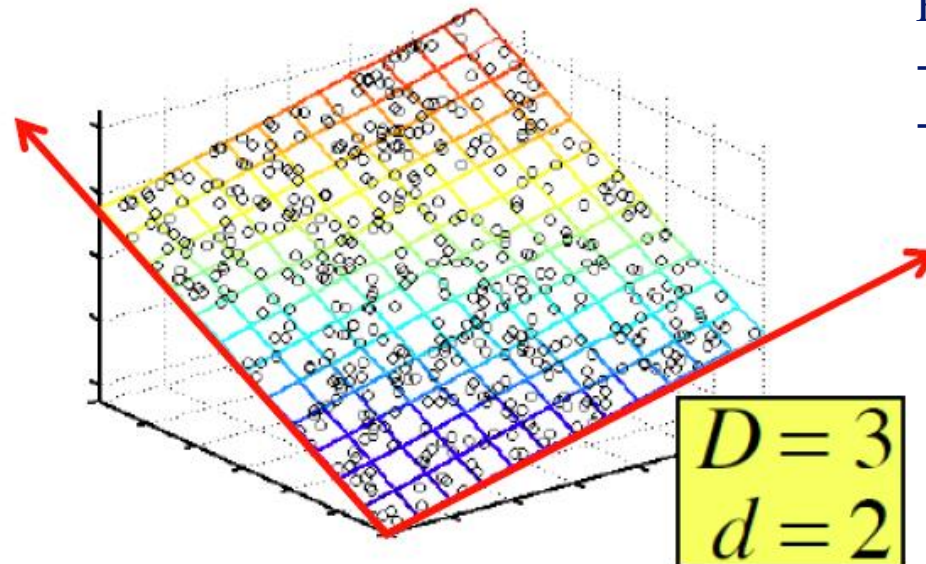
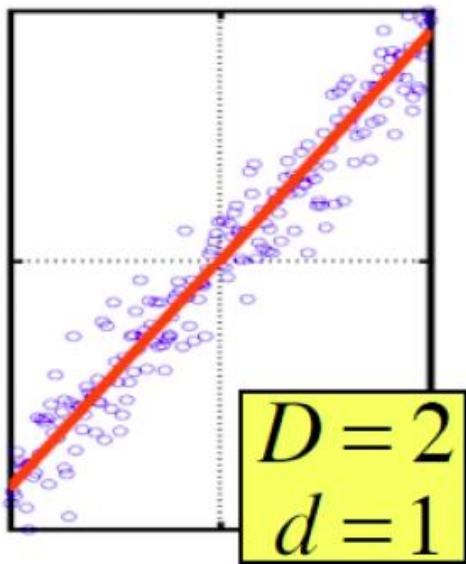
(B) 2-D



(C) 3-D

Dimensionality Reduction

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Benefits:

- Better visualization detects more easily patterns
- Faster and more efficient model

- **Assumption:** Data lies on or near a low d -dimensional subspace
- **Axes of this subspace are effective representation of the data**

Dimensionality Reduction

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- **Compress / reduce dimensionality:**

- 10^6 rows; 10^3 columns; no updates
- Random access to any cell(s); **small error: OK**

customer	day	We	Th	Fr	Sa	Su
		7/10/96	7/11/96	7/12/96	7/13/96	7/14/96
ABC Inc.		1	1	1	0	0
DEF Ltd.		2	2	2	0	0
GHI Inc.		1	1	1	0	0
KLM Co.		5	5	5	0	0
Smith		0	0	0	2	2
Johnson		0	0	0	3	3
Thompson		0	0	0	1	1

The above matrix is really “2-dimensional.” All rows can be reconstructed by scaling $[1 \ 1 \ 1 \ 0 \ 0]$ or $[0 \ 0 \ 0 \ 1 \ 1]$

Rank of a Matrix

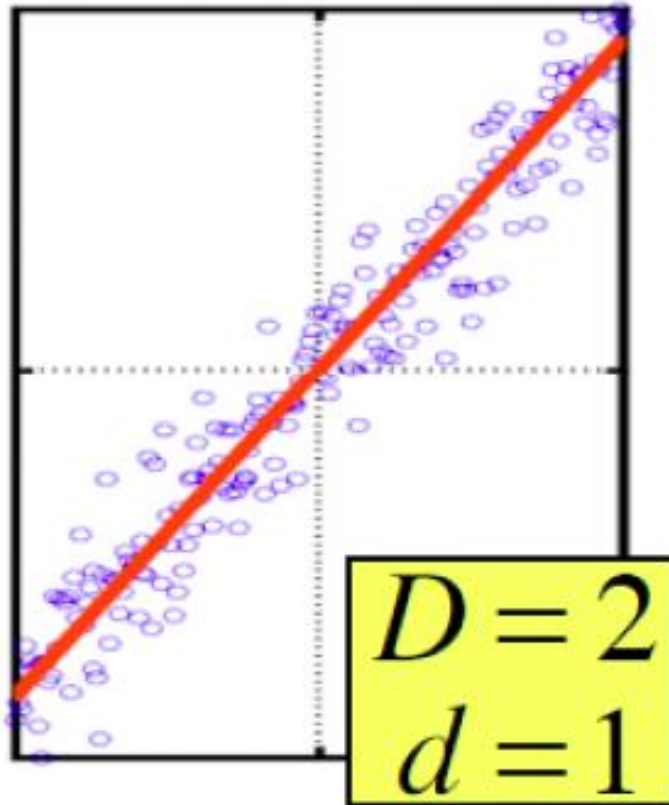
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- **Q:** What is **rank** of a matrix **A**?
- **A:** Number of **linearly independent** columns/rows of **A**
- **For example:**
 - Matrix $A = \begin{bmatrix} 1 & 2 & 1 \\ -2 & -3 & 1 \\ 3 & 5 & 0 \end{bmatrix}$ has rank **r=2**
 - **Why?** The first two rows are linearly independent, so the rank is at least 2, but all three rows are linearly dependent (the first is equal to the sum of the second and third: $r_1 = r_2 + r_3$, or $r_3 = r_1 - r_2$) so the rank must be less than 3.
- **Why do we care about low rank?**
 - We can write **A** as two “basis” vectors: $\begin{bmatrix} 1 & 2 & 1 \\ -2 & -3 & 1 \end{bmatrix}$
 - New coordinates: $\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$

Dimensionality Reduction

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- Goal of dimensionality reduction is to discover the axis of data!



Rather than representing every point with 2 coordinates we represent each point with 1 coordinate (corresponding to the position of the point on the red line).

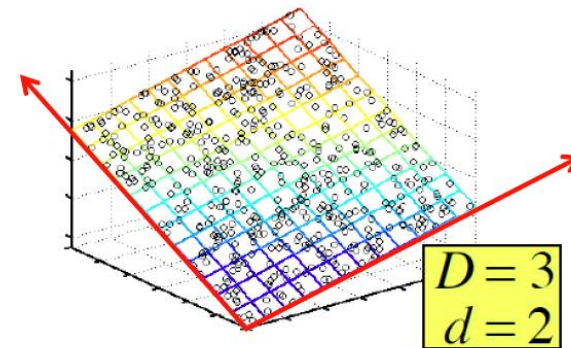
By doing this we incur a bit of **error** as the points do not exactly lie on the line

Why Reduce Dimensions?

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Why reduce dimensions?

- **Discover hidden correlations/topics**
 - Words that occur commonly together
- **Remove redundant and noisy features**
 - Not all words are useful
- **Interpretation and visualization**
- **Easier storage and processing of the data**



PCA

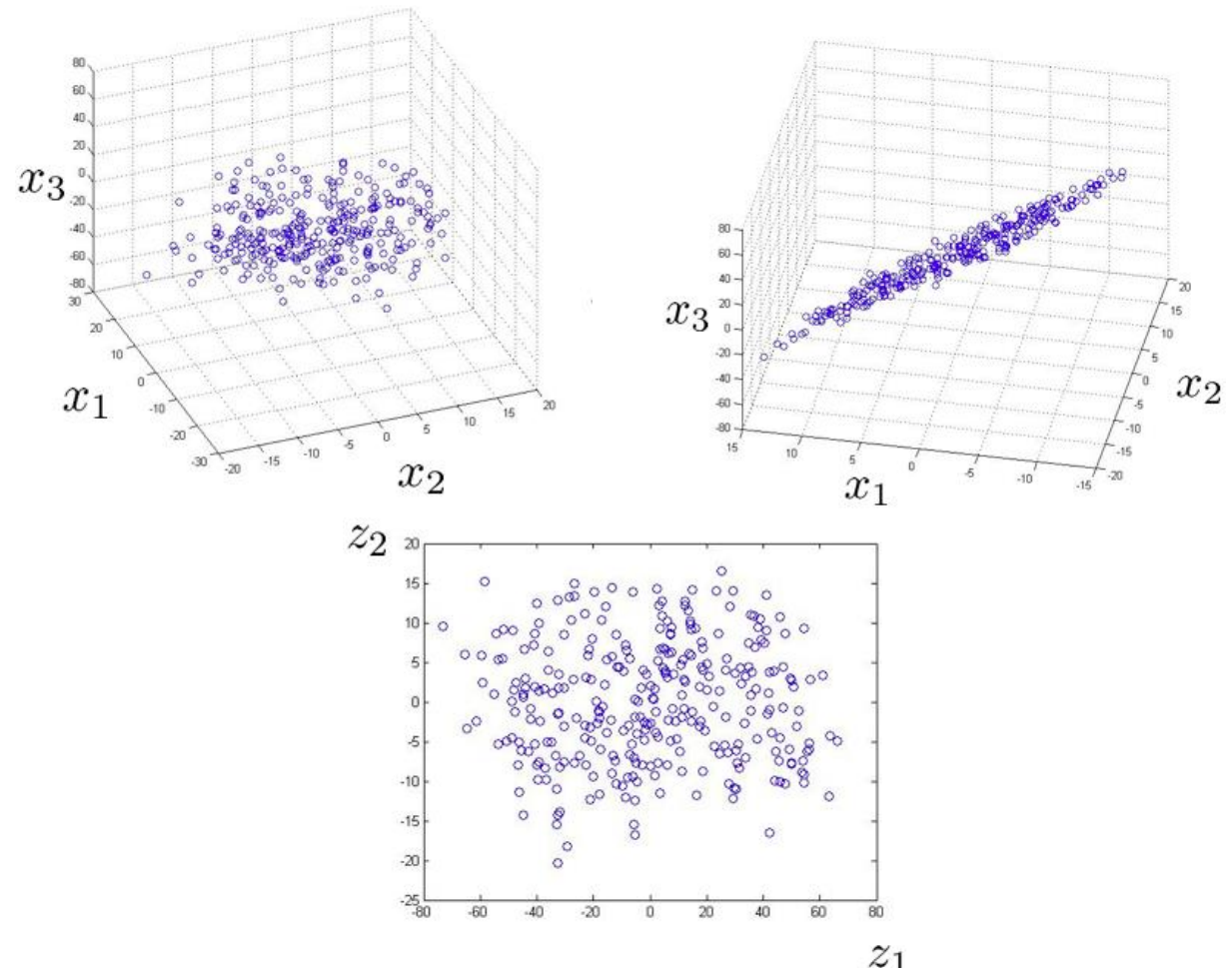
Principal Component Analysis



Principal Components Analysis (PCA)

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- An exploratory technique used to reduce the dimensionality of the data set to lower dimension
- Can be used to:
 - Reduce number of dimensions in data
 - Find patterns in high-dimensional data
 - Visualize data of high dimensionality
- Example applications:
 - Face recognition
 - Image compression
 - Gene expression analysis



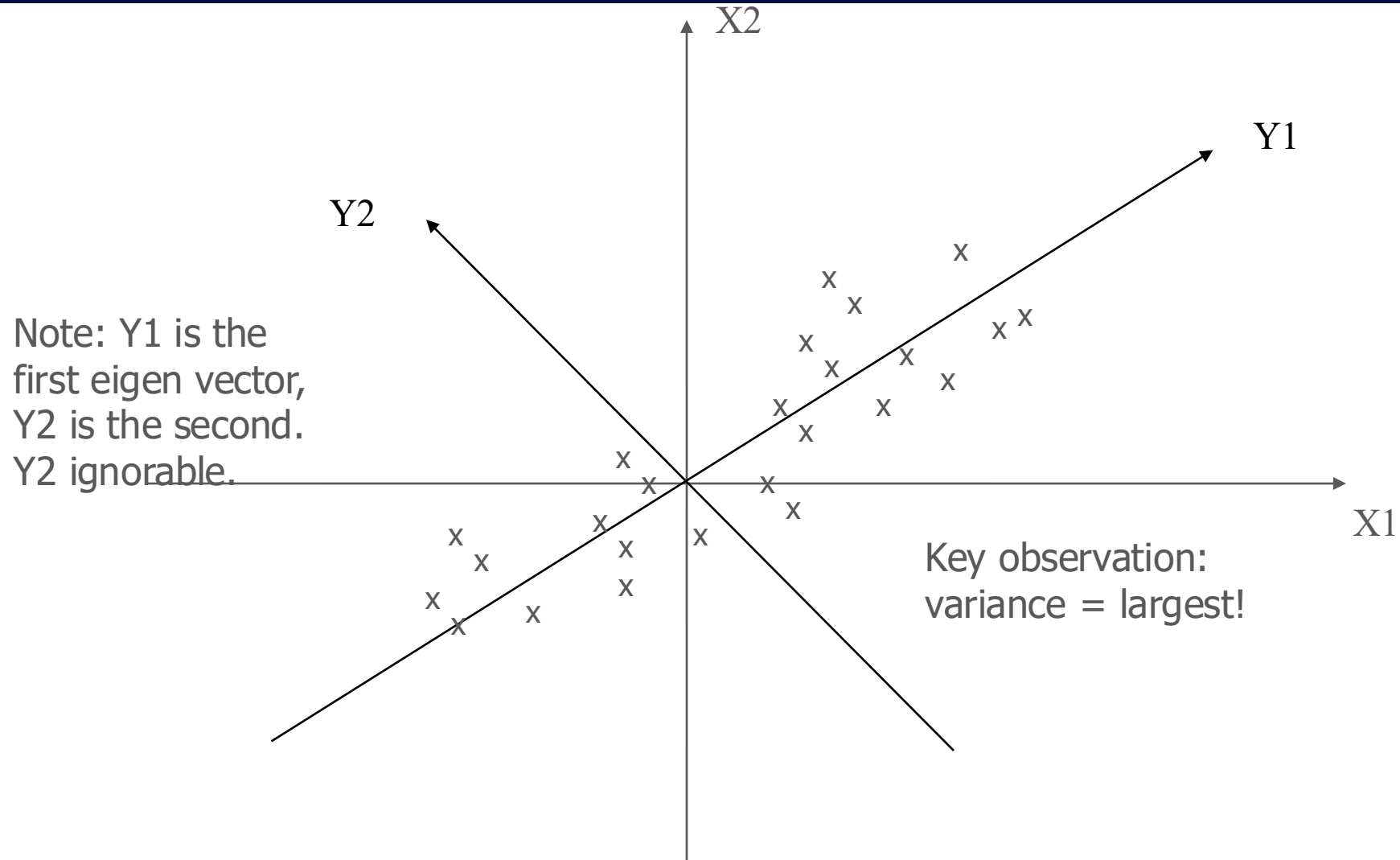
Principal Components Analysis Ideas

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- Does the data set ‘span’ the whole of d dimensional space?
- For a matrix of m samples $\times$ n genes, create a new covariance matrix of size $n \times n$.
- Transform some large number of variables into a smaller number of uncorrelated variables called principal components (PCs) developed to capture as much of the variation in data as possible

Principal component analysis

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Principal Component Analysis: one attribute first

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- Question: how much spread is in the data along the axis? (distance to the mean)
- Variance=Standard deviation²

$$s^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n - 1}$$

Temperature
42
40
24
30
15
18
15
30
15
30
35
30
40
20



Two dimensions

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Covariance: measures the correlation between X and Y

- $\text{cov}(X,Y)=0$: independent
- $\text{Cov}(X,Y)>0$: move same direction
- $\text{Cov}(X,Y)<0$: move opposite direction

$$\text{cov}(X,Y) = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{(n-1)}$$

How about 3 variables?

X=Temperature	Y=Humidity
40	90
40	90
40	90
30	90
15	70
15	70
15	70
30	90
15	70
30	70
30	70
30	90
40	70
30	90

PCA - Definition

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- M : square matrix, λ : a constant, e : non-zero column vector of rows as M .
- Vectors e having same direction as Ae is called *eigenvector* of A
- λ : eigenvalue of M , e : eigenvector of M if $Me = \lambda e$ (e having same direction as Me)
- Eigen vector must be unit vector, $\sum e_{ij}^2 = 1$, $e_{11} > 0$.

$$\begin{bmatrix} 3 & 2 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix} = 7 \begin{bmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix}$$

M Eigenvalue Eigenvector

Principal components

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1. principal component (PC1)

The eigenvalue with the largest absolute value will indicate that the data have the largest variance along its eigenvector (most "spread out")

- the direction along which there is greatest variation

2. principal component (PC2)

The direction with maximum variation left in data, orthogonal to the PC1.

- In general, only few directions manage to capture most of the variability in the data.

PCA: try to Project high-dimensional data onto the most important axes

Steps of PCA

- For matrix M , vectors $\mathbf{e}$ (=column vector) having same direction as $M\mathbf{e}$:
 - *eigenvectors* of M is $\mathbf{e}$ such that $M\mathbf{e}=\lambda\mathbf{e}$,
 - λ is called an *eigenvalue* of C .
- $M\mathbf{e}=\lambda\mathbf{e} \Leftrightarrow (M-\lambda I)\mathbf{e}=0$
- Calculate $\det(\mathbf{M}-\lambda I)$
- **Find** λ such that $\det(\mathbf{M}-\lambda I) = 0$
- Replace λ in equations from $M\mathbf{e}=\lambda\mathbf{e}$ to find $\mathbf{e}$
- Divide each component of $\mathbf{e}$ by **sum of squares of all components** to derive the unit vector

PCA example

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$$M = \begin{bmatrix} 3 & 2 \\ 2 & 6 \end{bmatrix} \quad M - \lambda I = \begin{bmatrix} 3 - \lambda & 2 \\ 2 & 6 - \lambda \end{bmatrix}$$

$$\det(M - \lambda I) = \lambda^2 - 9\lambda + 14$$
$$\det(.) = 0, \lambda = 7, \lambda = 2$$

$$e = \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\begin{bmatrix} 3 & 2 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 7 \begin{bmatrix} x \\ y \end{bmatrix}$$

$$Me = \lambda e$$

$$3x + 2y = 7x$$

$$2x + 6y = 7y$$

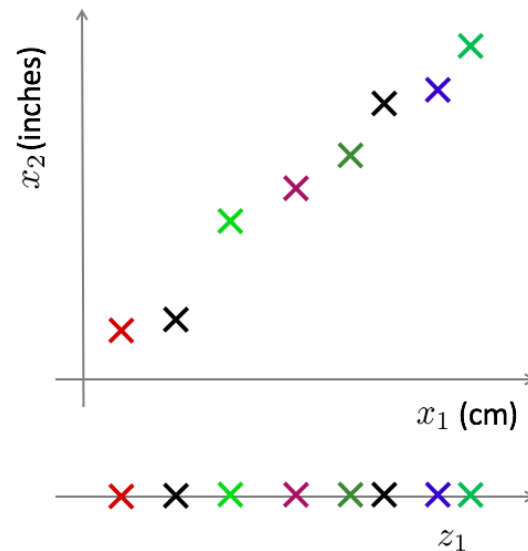
$$y = 2x$$

$$\text{Possible } e = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \text{Unit vector } e (\lambda = 7) = \begin{bmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix} \quad e (\lambda = 2) = \begin{bmatrix} 2/\sqrt{5} \\ -1/\sqrt{5} \end{bmatrix}$$

PCA in dimensionality reduction

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- Goals: Find k -dimension projection that best preserves variance
- Compute eigenvalues λ and eigenvectors $\mathbf{e}$ of $\mathbf{M}^T\mathbf{M}$
 - $\mathbf{E}$: matrix of eigenvectors - rigid rotation in a high dimensional space
- Select k first columns of $\mathbf{E}$ and project points onto subspace spanned $\mathbf{ME}_k$



Example

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$$\mathbf{M} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \\ 3 & 4 \\ 4 & 3 \end{bmatrix} \quad \mathbf{A} = \mathbf{M}^T \mathbf{M} = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 1 \\ 3 & 4 \\ 4 & 3 \end{bmatrix} = \begin{bmatrix} 30 & 28 \\ 28 & 30 \end{bmatrix}$$

$$(30 - \lambda)(30 - \lambda) - 28 \times 28 = 0 \quad \lambda = 58, \lambda = 2$$

$$\begin{bmatrix} 30 & 28 \\ 28 & 30 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 58 \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\mathbf{e}_1 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

$$\mathbf{E} = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

$$\begin{bmatrix} 30 & 28 \\ 28 & 30 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 2 \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\mathbf{e}_2 = \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

$$\mathbf{ME}_1 = \begin{bmatrix} 3/\sqrt{2} \\ 3/\sqrt{2} \\ 7/\sqrt{2} \\ 7/\sqrt{2} \end{bmatrix}$$

$$\mathbf{A}\mathbf{e} = \lambda\mathbf{e}$$

Power Iteration

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- Choose a random vector $\mathbf{x}_0$

- Calculate

$$\mathbf{x}_{k+1} := \frac{M\mathbf{x}_k}{\|M\mathbf{x}_k\|}$$

Frobenius norm:

$$\|M\|_F = \sqrt{\sum_{ij} M_{ij}^2}$$

- $\|N\|$: Frobenius norm – square root of the sum of square of the elements of N
- Multiply $\mathbf{x}_k$ until convergence $\|\mathbf{x}_k - \mathbf{x}_{k+1}\| < \text{threshold}$
- $\mathbf{x} = \text{final } \mathbf{x}_k$, $\mathbf{x}$ is approximately the principal eigenvector of M
- Compute $\lambda_1 = \mathbf{x}^T M \mathbf{x}$ ($M\mathbf{x} = \lambda\mathbf{x}$)

For second eigenpair, create $M^* = M - \lambda_1 \mathbf{x} \mathbf{x}^T$

- Apply Power Iteration on M^* to compute the second largest eigenvalue

Exercise

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$$M = \begin{bmatrix} 1 & 3 \\ 3 & 5 \\ 5 & 3 \\ 0 & 4 \\ 1 & 5 \end{bmatrix}$$

1. Calculate the eigenvectors and eigenvalues $M^T M$.
2. Reduce M to one dimension

SVD

Singular vector decomposition



SVD – Definition (Singular Value Decomposition)

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$$\mathbf{A}_{[m \times n]} = \mathbf{U}_{[m \times r]} \mathbf{\Sigma}_{[r \times r]} (\mathbf{V}_{[n \times r]})^T$$

- **A: Input data matrix**
 - $m \times n$ matrix (e.g., m documents, n terms)
- **U: Left singular vectors**
 - $m \times r$ matrix (m documents, r concepts)
- **Σ : Singular values**
 - $r \times r$ diagonal matrix (strength of each ‘concept’)
(r : rank of the matrix $\mathbf{A}$)
- **V: Right singular vectors**
 - $n \times r$ matrix (n terms, r concepts)

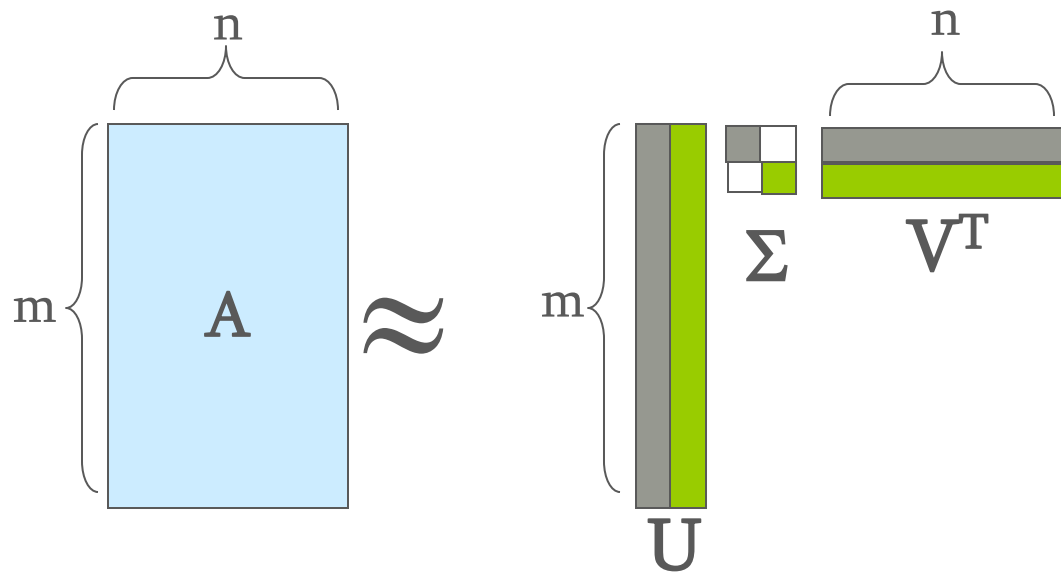
Produce a low-dimensional representation of a high-dimensional matrix

- eliminate the less important parts,
- produce an approximate representation with any desired number of dimensions

SVD

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$$\mathbf{A} \approx \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T = \sum_i \sigma_i \mathbf{u}_i \circ \mathbf{v}_i$$



$\mathbf{A} =$

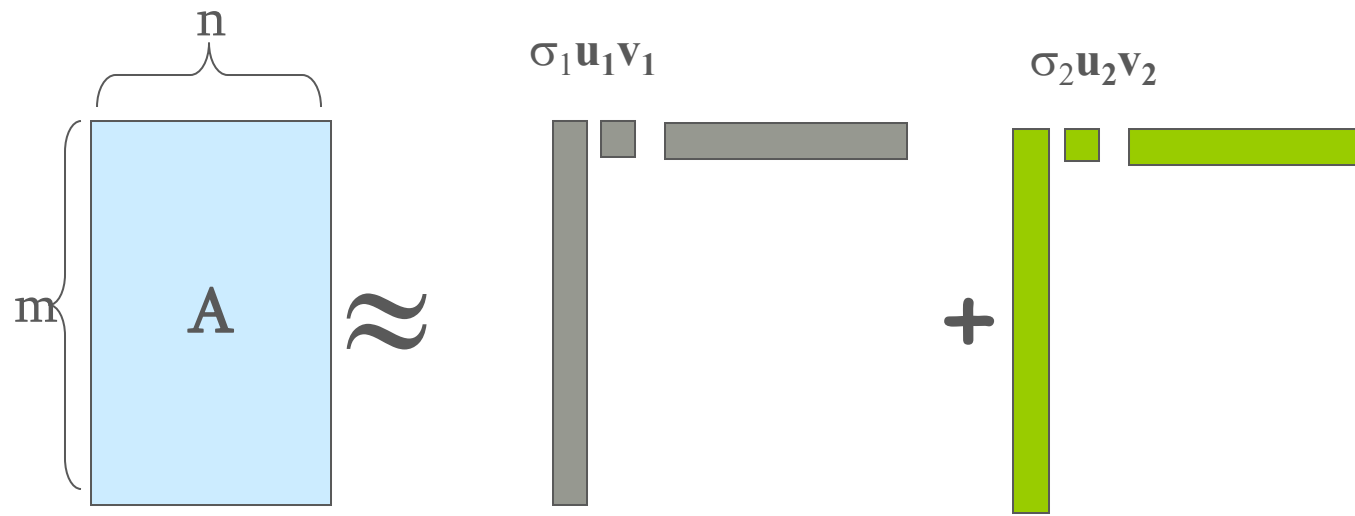
	Matrix	Alien	Star Wars	Casablanca	Titanic
Joe	1	1	1	0	0
Jim	3	3	3	0	0
John	4	4	4	0	0
Jack	5	5	5	0	0
Jill	0	0	0	4	4
Jenny	0	0	0	5	5
Jane	0	0	0	2	2

$$\approx \begin{bmatrix} .14 & 0 \\ .42 & 0 \\ .56 & 0 \\ .70 & 0 \\ 0 & .60 \\ 0 & .75 \\ 0 & .30 \end{bmatrix} \begin{bmatrix} 12.4 & 0 \\ 0 & 9.5 \end{bmatrix} \begin{bmatrix} .58 & .58 & .58 & 0 & 0 \\ 0 & 0 & 0 & .71 & .71 \end{bmatrix}$$

SVD

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$$\mathbf{A} \approx \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T = \sum_i \sigma_i \mathbf{u}_i \circ \mathbf{v}_i^T$$



σ_i ... scalar

$\mathbf{u}_i$... vector

$\mathbf{v}_i$... vector

SVD - Properties

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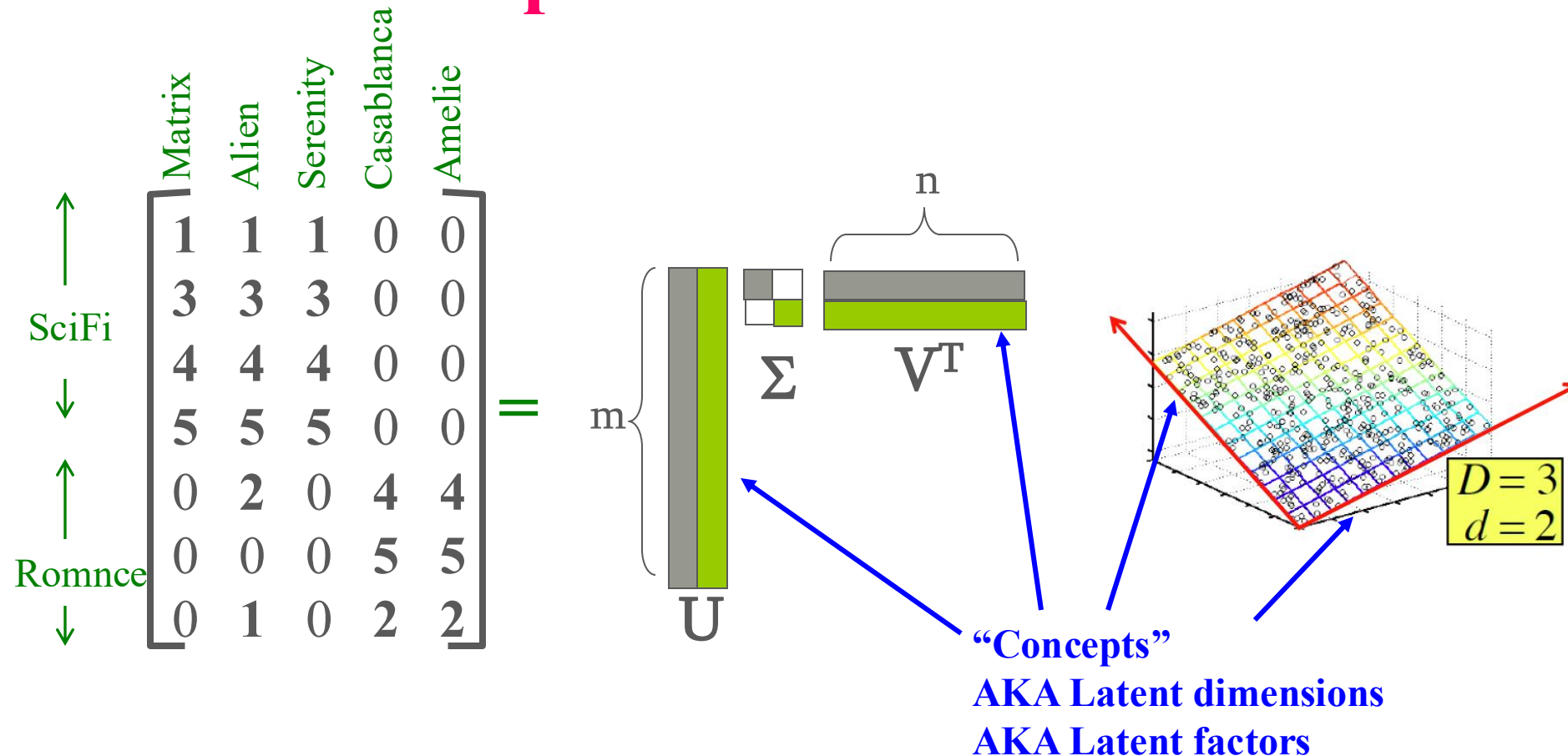
It is **always** possible to decompose a real matrix A into $A = U \Sigma V^T$, where

- U, Σ, V : **unique**
- U, V : **column orthonormal**
 - $U^T U = I; V^T V = I$ (I : identity matrix)
 - (Columns are orthogonal unit vectors)
- Σ : **diagonal**
 - Entries (**singular values**) are **positive**, and sorted in decreasing order ($\sigma_1 \geq \sigma_2 \geq \dots \geq 0$): express the important weights

SVD – Example: Users-to-Movies

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- $A = U \Sigma V^T$ - example: Users to Movies



SVD – Example: Users-to-Movies

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- $A = U \Sigma V^T$ - example: Users to Movies

$$\begin{array}{c}
 \begin{array}{c} \uparrow \\ \text{SciFi} \\ \downarrow \\ \uparrow \\ \text{Romnce} \\ \downarrow \end{array}
 \end{array}
 \begin{array}{c}
 \text{Matrix} \\
 \text{Alien} \\
 \text{Serenity} \\
 \text{Casablanca} \\
 \text{Amelie}
 \end{array}
 \begin{bmatrix}
 1 & 1 & 1 & 0 & 0 \\
 3 & 3 & 3 & 0 & 0 \\
 4 & 4 & 4 & 0 & 0 \\
 5 & 5 & 5 & 0 & 0 \\
 0 & 2 & 0 & 4 & 4 \\
 0 & 0 & 0 & 5 & 5 \\
 0 & 1 & 0 & 2 & 2
 \end{bmatrix}
 =
 \begin{bmatrix}
 0.13 & 0.02 & -0.01 \\
 0.41 & 0.07 & -0.03 \\
 0.55 & 0.09 & -0.04 \\
 0.68 & 0.11 & -0.05 \\
 0.15 & -0.59 & 0.65 \\
 0.07 & -0.73 & -0.67 \\
 0.07 & -0.29 & 0.32
 \end{bmatrix}
 \mathbf{X}
 \begin{bmatrix}
 12.4 & 0 & 0 \\
 0 & 9.5 & 0 \\
 0 & 0 & 1.3
 \end{bmatrix}
 \mathbf{X}
 \begin{bmatrix}
 0.56 & 0.59 & 0.56 & 0.09 & 0.09 \\
 0.12 & -0.02 & 0.12 & -0.69 & -0.69 \\
 0.40 & -0.80 & 0.40 & 0.09 & 0.09
 \end{bmatrix}$$

SVD – Example: Users-to-Movies

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- $A = U \Sigma V^T$ - example: Users to Movies

Diagram illustrating the SVD decomposition of a Users-to-Movies matrix A .

The matrix A (Users to Movies) is shown with rows representing movies and columns representing users. The rows are labeled: Matrix, Alien, Serenity, Casablanca, Amelie. The columns are labeled: SciFi, Romance, and an unlabeled column.

The matrix A is decomposed into three matrices: U , Σ , and V^T .

Matrix U (Users to Movies):

	Matrix	Alien	Serenity	Casablanca	Amelie
SciFi	1	1	1	0	0
	3	3	3	0	0
	4	4	4	0	0
	5	5	5	0	0
Romnce	0	2	0	4	4
	0	0	0	5	5
	0	1	0	2	2

Matrix Σ (Concepts):

	SciFi-concept	Romance-concept	
	0.13	0.02	-0.01
	0.41	0.07	-0.03
	0.55	0.09	-0.04
	0.68	0.11	-0.05
	0.15	-0.59	0.65
	0.07	-0.73	-0.67
	0.07	-0.29	0.32

Matrix V^T (Concepts):

	12.4	0	0
	0	9.5	0
	0	0	1.3

The decomposition is shown as:

$$A = U \Sigma V^T$$

The resulting matrix V^T is shown below:

0.56	0.59	0.56	0.09	0.09
0.12	-0.02	0.12	-0.69	-0.69
0.40	-0.80	0.40	0.09	0.09

SVD – Example: Users-to-Movies

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- $A = U \Sigma V^T$ - example:

U is “user-to-concept”
similarity matrix

$$\begin{array}{c}
 \begin{array}{c} \uparrow \\ \text{SciFi} \\ \downarrow \\ \uparrow \\ \text{Romnce} \\ \downarrow \end{array}
 \begin{array}{c} \text{Matrix} \\ \text{Alien} \\ \text{Serenity} \\ \text{Casablanca} \\ \text{Amelie} \end{array}
 \begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 3 & 3 & 3 & 0 & 0 \\ 4 & 4 & 4 & 0 & 0 \\ 5 & 5 & 5 & 0 & 0 \\ 0 & 2 & 0 & 4 & 4 \\ 0 & 0 & 0 & 5 & 5 \\ 0 & 1 & 0 & 2 & 2 \end{bmatrix}
 =
 \begin{array}{c} \text{SciFi-concept} \\ \text{Romance-concept} \end{array}
 \begin{bmatrix} 0.13 & 0.02 & -0.01 \\ 0.41 & 0.07 & -0.03 \\ 0.55 & 0.09 & -0.04 \\ 0.68 & 0.11 & -0.05 \\ 0.15 & -0.59 & 0.65 \\ 0.07 & -0.73 & -0.67 \\ 0.07 & -0.29 & 0.32 \end{bmatrix}
 \mathbf{X}
 \begin{bmatrix} 12.4 & 0 & 0 \\ 0 & 9.5 & 0 \\ 0 & 0 & 1.3 \end{bmatrix}
 \mathbf{X}
 \begin{bmatrix} 0.56 & 0.59 & 0.56 & 0.09 & 0.09 \\ 0.12 & -0.02 & 0.12 & -0.69 & -0.69 \\ 0.40 & -0.80 & 0.40 & 0.09 & 0.09 \end{bmatrix}
 \end{array}$$

SVD – Example: Users-to-Movies

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• $A = U \Sigma V^T$ - example:

Diagram illustrating the SVD decomposition of a user-movie rating matrix A into U , Σ , and V^T .

Matrix A (User-Movie Ratings):

	Matrix	Alien	Serenity	Casablanca	Amelie
SciFi	1	1	1	0	0
	3	3	3	0	0
	4	4	4	0	0
	5	5	5	0	0
	0	2	0	4	4
Romnce	0	0	0	5	5
	0	1	0	2	2

Matrix U (User-Concepts):

	SciFi-concept		
SciFi	0.13	0.02	-0.01
	0.41	0.07	-0.03
	0.55	0.09	-0.04
	0.68	0.11	-0.05
	0.15	-0.59	0.65
Romnce	0.07	-0.73	-0.67
	0.07	-0.29	0.32

Matrix Σ (Singular Values):

12.4	0	0
0	9.5	0
0	0	1.3

Matrix V^T (Concept-Movie):

0.56	0.59	0.56	0.09	0.09
0.12	-0.02	0.12	-0.69	-0.69
0.40	-0.80	0.40	0.09	0.09

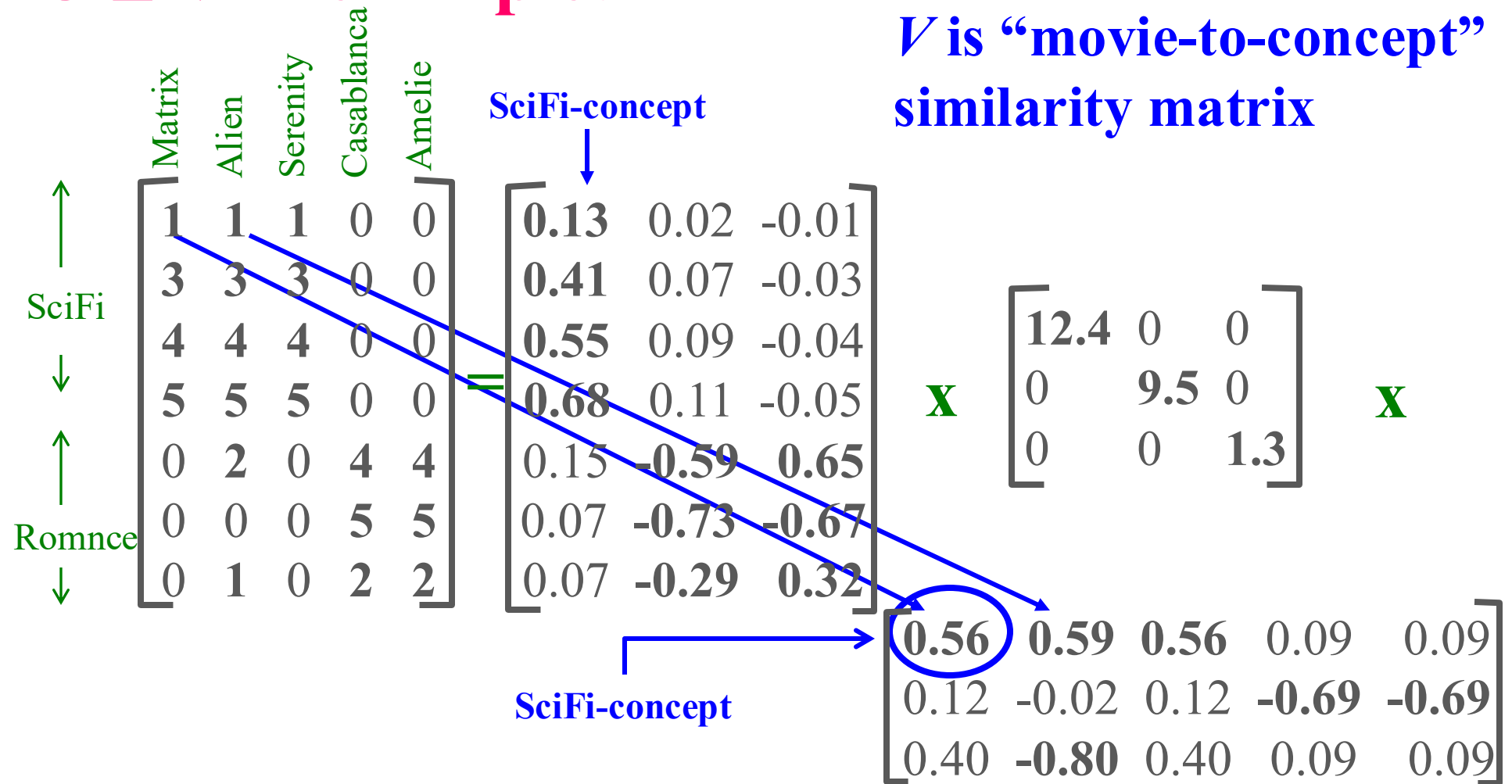
Annotations:

- Green arrows indicate the relationship between the SciFi and Romnce rows in matrix A and the corresponding rows in matrix U .
- A blue arrow points to the SciFi-concept column in matrix U .
- A blue circle highlights the value 12.4 in matrix Σ , labeled "strength" of the SciFi-concept.

SVD – Example: Users-to-Movies

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- $A = U \Sigma V^T$ - example:



SVD - Interpretation #1

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‘**movies**’, ‘**users**’ and ‘**concepts**’:

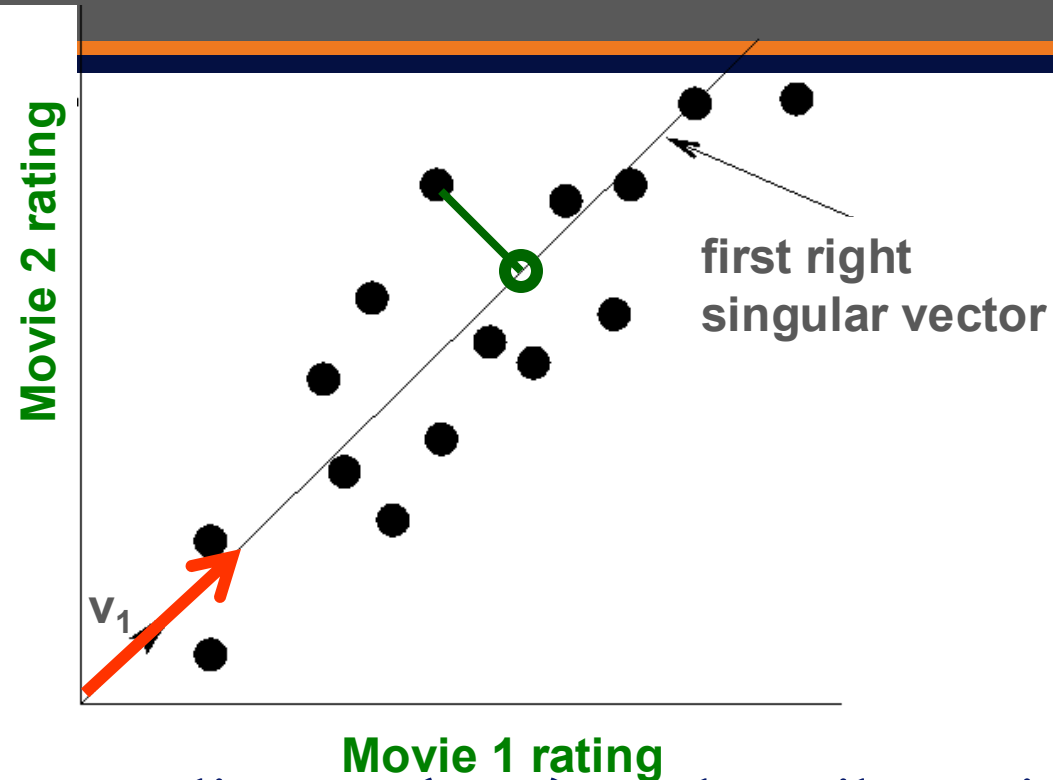
- U : user-to-concept similarity matrix
- V : movie-to-concept similarity matrix
- Σ : its diagonal elements: ‘strength’ of each concept

Dimensionality Reduction with SVD



SVD – Dimensionality Reduction

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- Instead of using two coordinates (x, y) to describe point locations, let's use only one coordinate (z)
- Point's position is its location along vector v_1
- How to choose v_1 ? **Minimize reconstruction error**

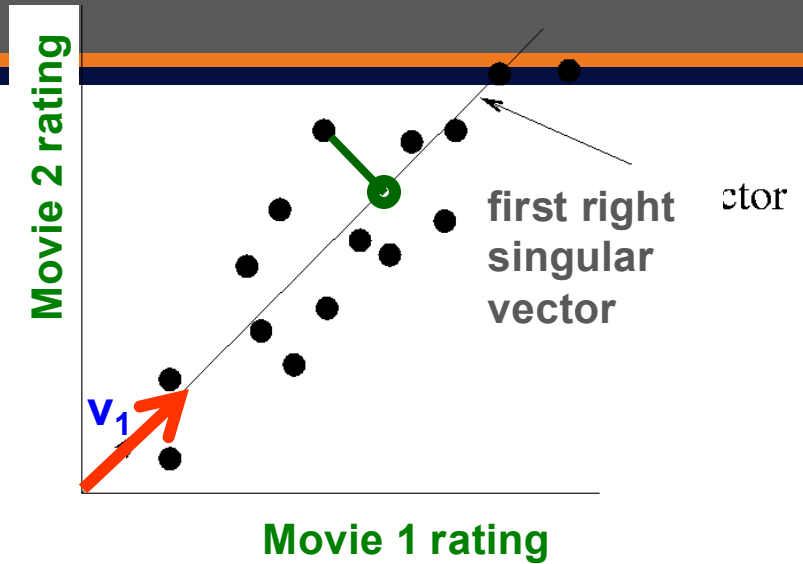
SVD – Dimensionality Reduction

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- **Goal:** Minimize the sum of reconstruction errors:

$$\sum_{i=1}^N \sum_{j=1}^D \|x_{ij} - z_{ij}\|^2$$

- where x_{ij} are the “old” and z_{ij} are the “new” coordinates
- **SVD gives ‘best’ axis to project on:**
 - ‘best’ = minimizing the reconstruction errors
- In other words, **minimum reconstruction error**



SVD - Interpretation #2

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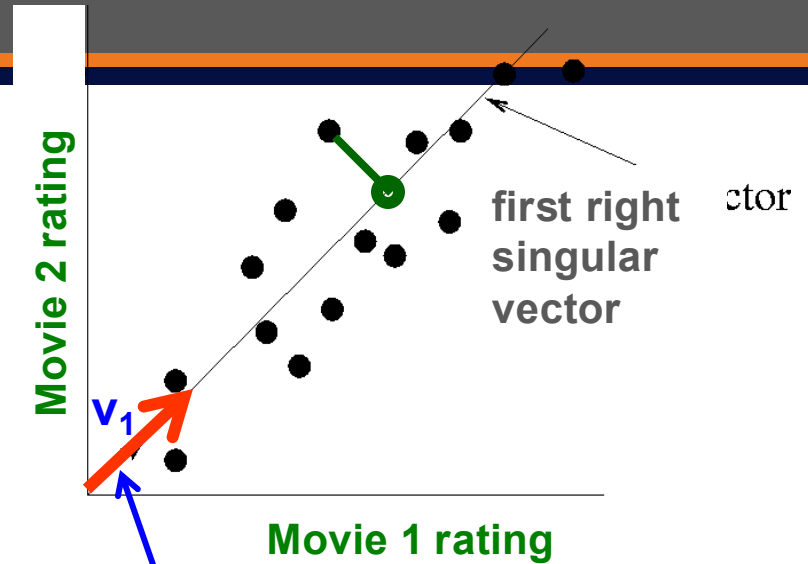
• $A = U \Sigma V^T$ - example:

- V : “movie-to-concept” matrix
- U : “user-to-concept” matrix

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 3 & 3 & 3 & 0 & 0 \\ 4 & 4 & 4 & 0 & 0 \\ 5 & 5 & 5 & 0 & 0 \\ 0 & 2 & 0 & 4 & 4 \\ 0 & 0 & 0 & 5 & 5 \\ 0 & 1 & 0 & 2 & 2 \end{bmatrix} = \begin{bmatrix} 0.13 & 0.02 & -0.01 \\ 0.41 & 0.07 & -0.03 \\ 0.55 & 0.09 & -0.04 \\ 0.68 & 0.11 & -0.05 \\ 0.15 & -0.59 & 0.65 \\ 0.07 & -0.73 & -0.67 \\ 0.07 & -0.29 & 0.32 \end{bmatrix} \mathbf{X}$$

$$\begin{bmatrix} 12.4 & 0 & 0 \\ 0 & 9.5 & 0 \\ 0 & 0 & 1.3 \end{bmatrix} \mathbf{X}$$

$$\begin{bmatrix} 0.56 & 0.59 & 0.56 & 0.09 & 0.09 \\ 0.12 & -0.02 & 0.12 & -0.69 & -0.69 \\ 0.40 & -0.80 & 0.40 & 0.09 & 0.09 \end{bmatrix}$$



SVD - Interpretation #2

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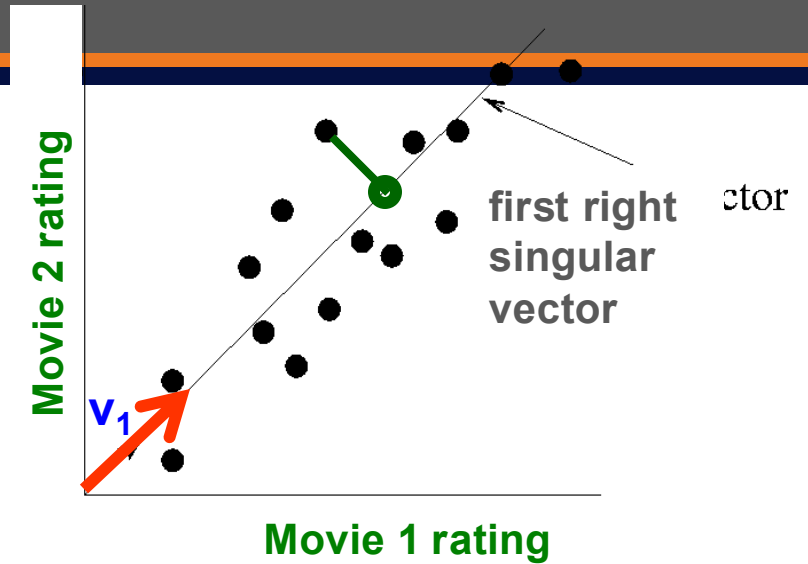
- $A = U \Sigma V^T$ - example:

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 3 & 3 & 3 & 0 & 0 \\ 4 & 4 & 4 & 0 & 0 \\ 5 & 5 & 5 & 0 & 0 \\ 0 & 2 & 0 & 4 & 4 \\ 0 & 0 & 0 & 5 & 5 \\ 0 & 1 & 0 & 2 & 2 \end{bmatrix} = \begin{bmatrix} 0.13 & 0.02 & -0.01 \\ 0.41 & 0.07 & -0.03 \\ 0.55 & 0.09 & -0.04 \\ 0.68 & 0.11 & -0.05 \\ 0.15 & -0.59 & 0.65 \\ 0.07 & -0.73 & -0.67 \\ 0.07 & -0.29 & 0.32 \end{bmatrix} \mathbf{X}$$

$$\begin{bmatrix} 12.4 & 0 & 0 \\ 0 & 9.5 & 0 \\ 0 & 0 & 1.3 \end{bmatrix} \mathbf{X}$$

$$\begin{bmatrix} 0.56 & 0.59 & 0.56 & 0.09 & 0.09 \\ 0.12 & -0.02 & 0.12 & -0.69 & -0.69 \\ 0.40 & -0.80 & 0.40 & 0.09 & 0.09 \end{bmatrix}$$

variance ('spread')
on the v_1 axis



SVD - Interpretation #2

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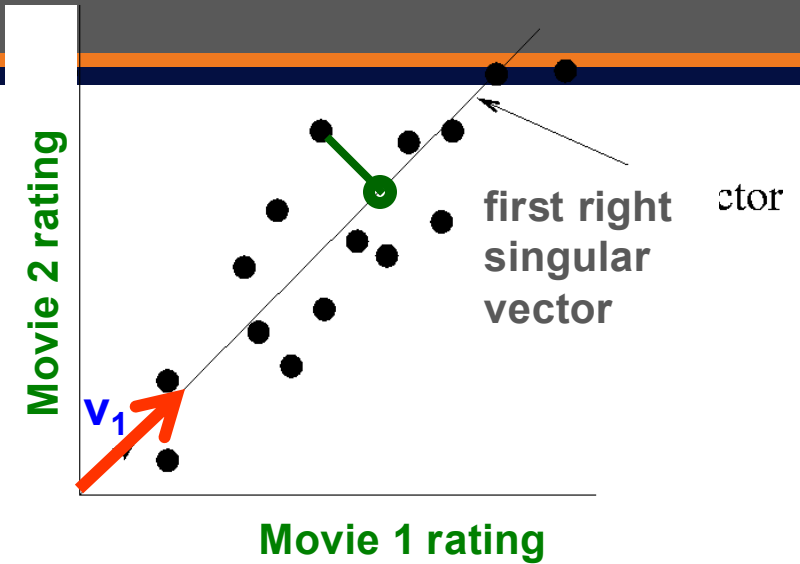
$A = U \Sigma V^T$ - example:

- $U \Sigma$: Gives the coordinates of the points in the projection axis

1	1	1	0	0
3	3	3	0	0
4	4	4	0	0
5	5	5	0	0
0	2	0	4	4
0	0	0	5	5
0	1	0	2	2

Projection of users
on the “Sci-Fi” axis
($U \Sigma$)^T:

1.61	0.19	-0.01
5.08	0.66	-0.03
6.82	0.85	-0.05
8.43	1.04	-0.06
1.86	-5.60	0.84
0.86	-6.93	-0.87
0.86	-2.75	0.41



SVD - Interpretation #2

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More details

- **Q:** How exactly is dim. reduction done?

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 3 & 3 & 3 & 0 & 0 \\ 4 & 4 & 4 & 0 & 0 \\ 5 & 5 & 5 & 0 & 0 \\ 0 & 2 & 0 & 4 & 4 \\ 0 & 0 & 0 & 5 & 5 \\ 0 & 1 & 0 & 2 & 2 \end{bmatrix} = \begin{bmatrix} 0.13 & 0.02 & -0.01 \\ 0.41 & 0.07 & -0.03 \\ 0.55 & 0.09 & -0.04 \\ 0.68 & 0.11 & -0.05 \\ 0.15 & -0.59 & 0.65 \\ 0.07 & -0.73 & -0.67 \\ 0.07 & -0.29 & 0.32 \end{bmatrix} \mathbf{x} \begin{bmatrix} 12.4 & 0 & 0 \\ 0 & 9.5 & 0 \\ 0 & 0 & 1.3 \end{bmatrix} \mathbf{x} \begin{bmatrix} 0.56 & 0.59 & 0.56 & 0.09 & 0.09 \\ 0.12 & -0.02 & 0.12 & -0.69 & -0.69 \\ 0.40 & -0.80 & 0.40 & 0.09 & 0.09 \end{bmatrix}$$

SVD - Interpretation #2

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More details

- **Q:** How exactly is dim. reduction done?
- **A:** Set smallest singular values to zero

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 3 & 3 & 3 & 0 & 0 \\ 4 & 4 & 4 & 0 & 0 \\ 5 & 5 & 5 & 0 & 0 \\ 0 & 2 & 0 & 4 & 4 \\ 0 & 0 & 0 & 5 & 5 \\ 0 & 1 & 0 & 2 & 2 \end{bmatrix} = \begin{bmatrix} 0.13 & 0.02 & -0.01 \\ 0.41 & 0.07 & -0.03 \\ 0.55 & 0.09 & -0.04 \\ 0.68 & 0.11 & -0.05 \\ 0.15 & -0.59 & 0.65 \\ 0.07 & -0.73 & -0.67 \\ 0.07 & -0.29 & 0.32 \end{bmatrix} \mathbf{X} \begin{bmatrix} 12.4 & 0 & 0 \\ 0 & 9.5 & 0 \\ 0 & 0 & \cancel{1.3} \end{bmatrix} \mathbf{X} \begin{bmatrix} 0.56 & 0.59 & 0.56 & 0.09 & 0.09 \\ 0.12 & -0.02 & 0.12 & -0.69 & -0.69 \\ 0.40 & -0.80 & 0.40 & 0.09 & 0.09 \end{bmatrix}$$

SVD - Interpretation #2

43

More details

- **Q:** How exactly is dim. reduction done?
- **A:** Set smallest singular values to zero

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 3 & 3 & 3 & 0 & 0 \\ 4 & 4 & 4 & 0 & 0 \\ 5 & 5 & 5 & 0 & 0 \\ 0 & 2 & 0 & 4 & 4 \\ 0 & 0 & 0 & 5 & 5 \\ 0 & 1 & 0 & 2 & 2 \end{bmatrix} \approx \begin{bmatrix} 0.13 & 0.02 & -0.01 \\ 0.41 & 0.07 & -0.03 \\ 0.55 & 0.09 & -0.04 \\ 0.68 & 0.11 & -0.05 \\ 0.15 & -0.59 & 0.65 \\ 0.07 & -0.73 & -0.67 \\ 0.07 & -0.29 & 0.32 \end{bmatrix} \mathbf{X} \begin{bmatrix} 12.4 & 0 & 0 \\ 0 & 9.5 & 0 \\ 0 & 0 & \cancel{1.3} \end{bmatrix} \mathbf{X} \begin{bmatrix} 0.56 & 0.59 & 0.56 & 0.09 & 0.09 \\ 0.12 & -0.02 & 0.12 & -0.69 & -0.69 \\ 0.40 & -0.80 & 0.40 & 0.09 & 0.09 \end{bmatrix}$$

SVD - Interpretation #2

44

More details

- **Q:** How exactly is dim. reduction done?
- **A:** Set smallest singular values to zero

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 3 & 3 & 3 & 0 & 0 \\ 4 & 4 & 4 & 0 & 0 \\ 5 & 5 & 5 & 0 & 0 \\ 0 & 2 & 0 & 4 & 4 \\ 0 & 0 & 0 & 5 & 5 \\ 0 & 1 & 0 & 2 & 2 \end{bmatrix} \approx \begin{bmatrix} 0.13 & 0.02 & -0.01 \\ 0.41 & 0.07 & -0.03 \\ 0.55 & 0.09 & -0.04 \\ 0.68 & 0.11 & -0.05 \\ 0.15 & -0.59 & 0.65 \\ 0.07 & -0.73 & -0.67 \\ 0.07 & -0.29 & 0.32 \end{bmatrix} \mathbf{X} \begin{bmatrix} 12.4 & 0 & 0 \\ 0 & 9.5 & 0 \\ 0 & 0 & 1.3 \end{bmatrix} \mathbf{X} \begin{bmatrix} 0.56 & 0.59 & 0.56 & 0.09 & 0.09 \\ 0.12 & -0.02 & 0.12 & -0.69 & -0.69 \\ 0.40 & -0.80 & 0.40 & 0.09 & 0.09 \end{bmatrix}$$

SVD - Interpretation #2

45

More details

- **Q:** How exactly is dim. reduction done?
- **A:** Set smallest singular values to zero

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 3 & 3 & 3 & 0 & 0 \\ 4 & 4 & 4 & 0 & 0 \\ 5 & 5 & 5 & 0 & 0 \\ 0 & 2 & 0 & 4 & 4 \\ 0 & 0 & 0 & 5 & 5 \\ 0 & 1 & 0 & 2 & 2 \end{bmatrix} \approx \begin{bmatrix} 0.13 & 0.02 \\ 0.41 & 0.07 \\ 0.55 & 0.09 \\ 0.68 & 0.11 \\ 0.15 & -0.59 \\ 0.07 & -0.73 \\ 0.07 & -0.29 \end{bmatrix} \mathbf{X} \begin{bmatrix} 12.4 & 0 \\ 0 & 9.5 \end{bmatrix} \mathbf{X} \begin{bmatrix} 0.56 & 0.59 & 0.56 & 0.09 & 0.09 \\ 0.12 & -0.02 & 0.12 & -0.69 & -0.69 \end{bmatrix}$$

SVD - Interpretation #2

46

More details

- **Q:** How exactly is dim. reduction done?
- **A:** Set smallest singular values to zero

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 3 & 3 & 3 & 0 & 0 \\ 4 & 4 & 4 & 0 & 0 \\ 5 & 5 & 5 & 0 & 0 \\ 0 & 2 & 0 & 4 & 4 \\ 0 & 0 & 0 & 5 & 5 \\ 0 & 1 & 0 & 2 & 2 \end{bmatrix} \approx \begin{bmatrix} 0.92 & 0.95 & 0.92 & 0.01 & 0.01 \\ 2.91 & 3.01 & 2.91 & -0.01 & -0.01 \\ 3.90 & 4.04 & 3.90 & 0.01 & 0.01 \\ 4.82 & 5.00 & 4.82 & 0.03 & 0.03 \\ 0.70 & 0.53 & 0.70 & 4.11 & 4.11 \\ -0.69 & 1.34 & -0.69 & 4.78 & 4.78 \\ 0.32 & 0.23 & 0.32 & 2.01 & 2.01 \end{bmatrix}$$

Frobenius norm:

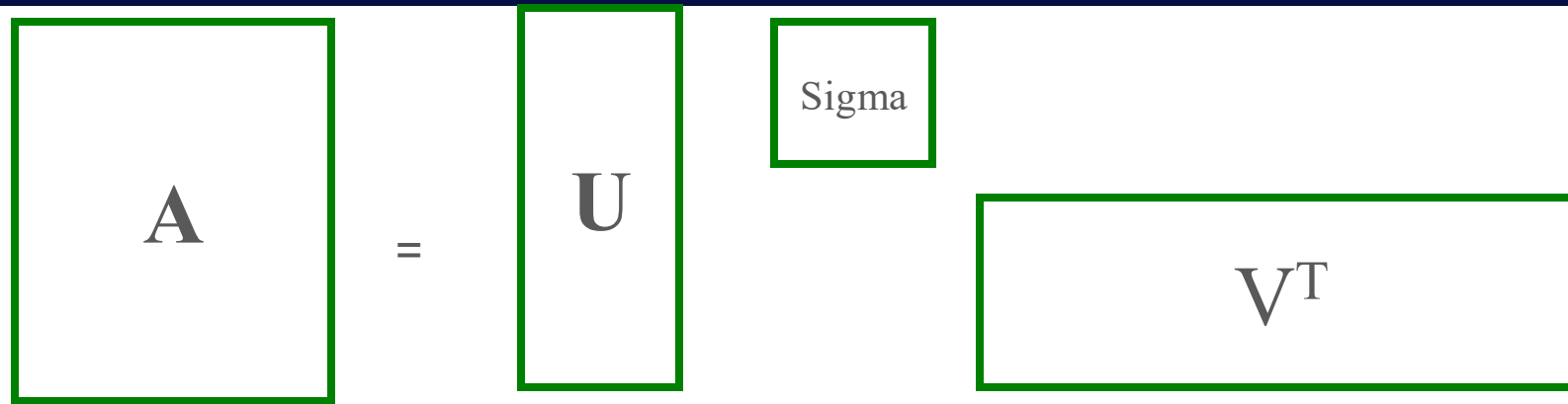
$$\|M\|_F = \sqrt{\sum_{ij} M_{ij}^2}$$

$$\|A-B\|_F = \sqrt{\sum_{ij} (A_{ij}-B_{ij})^2}$$

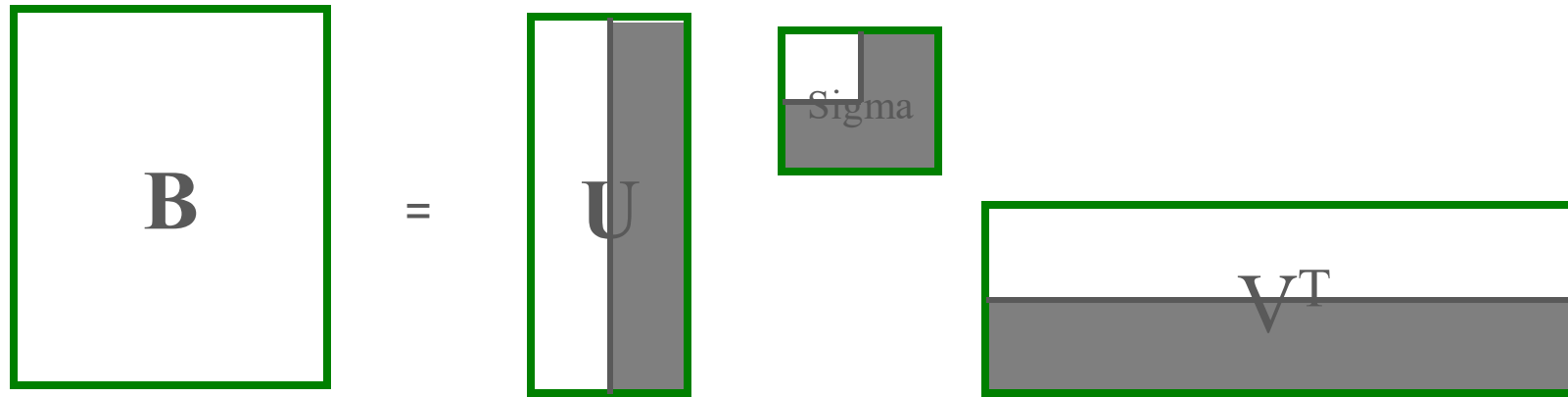
is "small"

SVD – Best Low Rank Approx.

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B is best approximation of A



SVD – Best Low Rank Approx.

48

- Theorem:**

Let $\mathbf{A} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^T$ and $\mathbf{B} = \mathbf{U} \mathbf{S} \mathbf{V}^T$ where

$\mathbf{S} = \text{diagonal } r \times r \text{ matrix}$ with $s_i = \sigma_i$ ($i = 1 \dots k$) else $s_i = 0$

then $\mathbf{B}$ is a **best rank($\mathbf{B}$)= k approx. to $\mathbf{A}$**

What do we mean by “best”:

- $\mathbf{B}$ is a solution to $\min_{\mathbf{B}} \|\mathbf{A} - \mathbf{B}\|_F$ where $\text{rank}(\mathbf{B}) = k$

$$\begin{pmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ x_{21} & x_{22} & \dots & \\ \vdots & \vdots & \ddots & \\ x_{m1} & & & x_{mn} \end{pmatrix}_{m \times n} = \begin{pmatrix} u_{11} & \dots & & \\ \vdots & \ddots & & \\ u_{m1} & & & \end{pmatrix}_{m \times r} \begin{pmatrix} \sigma_{11} & 0 & \dots \\ 0 & \ddots & \\ \vdots & & \end{pmatrix}_{r \times r} \begin{pmatrix} v_{11} & \dots & v_{1n} \\ \vdots & \ddots & \\ & & \end{pmatrix}_{r \times n}$$

$$\|\mathbf{A} - \mathbf{B}\|_F = \sqrt{\sum_{ij} (A_{ij} - B_{ij})^2}$$

SVD – Best Low Rank Approx.

- $\mathbf{A} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^T$, $\mathbf{B} = \mathbf{U} \mathbf{S} \mathbf{V}^T$ ($\sigma_1 \geq \sigma_2 \geq \dots \geq 0$, $\text{rank}(\mathbf{A})=r$)
- $\mathbf{S}$ = diagonal $n \times n$ matrix where $s_i = \sigma_i$ ($i=1 \dots k$) else $s_i=0$

then $\mathbf{B}$ is solution to $\min_{\mathbf{B}} \|\mathbf{A} - \mathbf{B}\|_F$, $\text{rank}(\mathbf{B})=k$

■ Why?

$$\min_{\mathbf{B}, \text{rank}(\mathbf{B})=k} \|\mathbf{A} - \mathbf{B}\|_F = \min \|\mathbf{\Sigma} - \mathbf{S}\|_F = \min_{s_i} \sum_{i=1}^r (\sigma_i - s_i)^2$$

$$\mathbf{U} \mathbf{\Sigma} \mathbf{V}^T - \mathbf{U} \mathbf{S} \mathbf{V}^T = \mathbf{U} (\mathbf{\Sigma} - \mathbf{S}) \mathbf{V}^T$$

- We want to choose s_i to minimize $\sum_i (\sigma_i - s_i)^2$
- Solution is to set $s_i = \sigma_i$ ($i=1 \dots k$) and other $s_i=0$

$$= \min_{s_i} \sum_{i=1}^k (\sigma_i - s_i)^2 + \sum_{i=k+1}^r \sigma_i^2 = \sum_{i=k+1}^r \sigma_i^2$$

SVD - Interpretation #2

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Equivalent:

‘spectral decomposition’ of the matrix:

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 3 & 3 & 3 & 0 & 0 \\ 4 & 4 & 4 & 0 & 0 \\ 5 & 5 & 5 & 0 & 0 \\ 0 & 2 & 0 & 4 & 4 \\ 0 & 0 & 0 & 5 & 5 \\ 0 & 1 & 0 & 2 & 2 \end{bmatrix} = \begin{bmatrix} | & | \\ u_1 & u_2 \\ | & | \end{bmatrix} \times \begin{bmatrix} \sigma_1 & \circ \\ \circ & \sigma_2 \end{bmatrix} \times \begin{bmatrix} \text{---} & v_1 & \text{---} \\ \text{---} & v_2 & \text{---} \end{bmatrix}$$

SVD - Interpretation #2

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Equivalent: ‘spectral decomposition’ of the matrix

$$\begin{array}{c} \updownarrow \\ n \end{array} \begin{array}{c} \leftarrow m \rightarrow \\ \begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 3 & 3 & 3 & 0 & 0 \\ 4 & 4 & 4 & 0 & 0 \\ 5 & 5 & 5 & 0 & 0 \\ 0 & 2 & 0 & 4 & 4 \\ 0 & 0 & 0 & 5 & 5 \\ 0 & 1 & 0 & 2 & 2 \end{bmatrix} \end{array} = \begin{array}{c} \leftarrow k \text{ terms} \rightarrow \\ \sigma_1 \begin{array}{c} \nearrow u_1 \\ n \times 1 \end{array} \begin{array}{c} \nwarrow v_1^T \\ 1 \times m \end{array} + \sigma_2 u_2 v_2^T + \dots \end{array}$$

Assume: $\sigma_1 \geq \sigma_2 \geq \sigma_3 \geq \dots \geq 0$

Why is setting small σ_i to 0 the right thing to do?

Vectors u_i and v_i are unit length, so σ_i scales them.

So, zeroing small σ_i introduces less error.

SVD - Interpretation #2

53

Q: How many σ s to keep?

A: Rule-of-a thumb:

keep 80-90% of 'energy' = $\sum_i \sigma_i^2$

$$\begin{array}{c} \updownarrow \\ \text{ } \\ \text{ } \\ \text{ } \\ \text{ } \\ \text{ } \\ \text{ } \\ \text{ } \\ \text{ } \\ \text{ } \\ \downarrow \\ \text{ } \end{array} \begin{array}{c} \leftarrow \text{ } m \text{ } \rightarrow \\ \left[\begin{array}{ccccc} 1 & 1 & 1 & 0 & 0 \\ 3 & 3 & 3 & 0 & 0 \\ 4 & 4 & 4 & 0 & 0 \\ 5 & 5 & 5 & 0 & 0 \\ 0 & 2 & 0 & 4 & 4 \\ 0 & 0 & 0 & 5 & 5 \\ 0 & 1 & 0 & 2 & 2 \end{array} \right] \end{array} = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T + \sigma_2 \mathbf{u}_2 \mathbf{v}_2^T + \dots$$

Assume: $\sigma_1 \geq \sigma_2 \geq \sigma_3 \geq \dots$

SVD - Complexity

54

- **To compute SVD:**
 - $O(nm^2)$ or $O(n^2m)$ (whichever is less)
- **But:**
 - Less work, if we just want singular values
 - or if we want first k singular vectors
 - or if the matrix is sparse

SVD - Conclusions so far

55

- **SVD:** $A = U \Sigma V^T$: **unique**
 - U : user-to-concept similarities
 - V : movie-to-concept similarities
 - Σ : strength of each concept
- **Dimensionality reduction:**
 - keep the few largest singular values (80-90% of 'energy')

Relation to Eigen-decomposition

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- **SVD gives us:**

- $A = U \Sigma V^T$

- **Eigen-decomposition:**

- $A = X \Lambda X^T$

- A is symmetric

- U, V, X are orthonormal ($U^T U = I$),

- Λ, Σ are diagonal

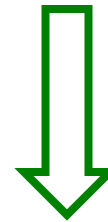
- **Now let's calculate:**

- $AA^T = U \Sigma V^T (U \Sigma V^T)^T = U \Sigma V^T (V \Sigma^T U^T) = U \Sigma \Sigma^T U^T$

- $A^T A = V \Sigma^T U^T (U \Sigma V^T) = V \Sigma \Sigma^T V^T$

$$\begin{matrix} \uparrow & \uparrow & \uparrow \\ X & \Lambda^2 & X^T \end{matrix}$$

Shows how to compute
SVD using eigenvalue
decomposition!



$$X \Lambda^2 X^T$$



Example of SVD & Conclusion



Case study: How to query?

58

- **Q: Find users that like 'Matrix'**
- **A: Map query into a 'concept space' – how?**

$$\begin{array}{c}
 \begin{array}{c} \uparrow \\ \text{SciFi} \\ \downarrow \\ \uparrow \\ \text{Romnce} \\ \downarrow \end{array}
 \begin{bmatrix}
 \text{Matrix} & \text{Alien} & \text{Serenity} & \text{Casablanca} & \text{Amelie} \\
 1 & 1 & 1 & 0 & 0 \\
 3 & 3 & 3 & 0 & 0 \\
 4 & 4 & 4 & 0 & 0 \\
 5 & 5 & 5 & 0 & 0 \\
 0 & 2 & 0 & 4 & 4 \\
 0 & 0 & 0 & 5 & 5 \\
 0 & 1 & 0 & 2 & 2
 \end{bmatrix}
 =
 \begin{bmatrix}
 0.13 & 0.02 & -0.01 \\
 0.41 & 0.07 & -0.03 \\
 0.55 & 0.09 & -0.04 \\
 0.68 & 0.11 & -0.05 \\
 0.15 & -0.59 & 0.65 \\
 0.07 & -0.73 & -0.67 \\
 0.07 & -0.29 & 0.32
 \end{bmatrix}
 \mathbf{X}
 \begin{bmatrix}
 12.4 & 0 & 0 \\
 0 & 9.5 & 0 \\
 0 & 0 & 1.3
 \end{bmatrix}
 \mathbf{X}
 \begin{bmatrix}
 0.56 & 0.59 & 0.56 & 0.09 & 0.09 \\
 0.12 & -0.02 & 0.12 & -0.69 & -0.69 \\
 0.40 & -0.80 & 0.40 & 0.09 & 0.09
 \end{bmatrix}
 \end{array}$$

Case study: How to query?

59

- **Q: Find users that like 'Matrix'**
- **A: Map query into a 'concept space' – how?**

$$\mathbf{q} = \begin{bmatrix} \text{Matrix} \\ 5 \\ \text{Alien} \\ 0 \\ \text{Serenity} \\ 0 \\ \text{Casablanca} \\ 0 \\ \text{Amelie} \\ 0 \end{bmatrix}$$

Project into concept space:

Inner product with each
'concept' vector $\mathbf{v}_i$

Case study: How to query?

60

Compactly, we have:

$$\mathbf{q}_{\text{concept}} = \mathbf{q} \mathbf{V}$$

E.g.:

$$\mathbf{q} = \begin{matrix} & \begin{matrix} \text{Matrix} \\ \text{Alien} \\ \text{Serenity} \\ \text{Casablanca} \\ \text{Amelie} \end{matrix} \\ \begin{matrix} 5 \\ 0 \\ 0 \\ 0 \\ 0 \end{matrix} & \end{matrix} \mathbf{x} \begin{matrix} \begin{bmatrix} 0.56 & 0.12 \\ 0.59 & -0.02 \\ 0.56 & 0.12 \\ 0.09 & -0.69 \\ 0.09 & -0.69 \end{bmatrix} \\ \text{movie-to-concept} \\ \text{similarities (V)} \end{matrix} = \begin{matrix} \text{SciFi-concept} \\ \downarrow \\ \begin{bmatrix} 2.8 & 0.6 \end{bmatrix} \end{matrix}$$

Case study: How to query?

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- How would the user d that rated ('Alien', 'Serenity') be handled?

$$\mathbf{d}_{\text{concept}} = \mathbf{d} \mathbf{V}$$

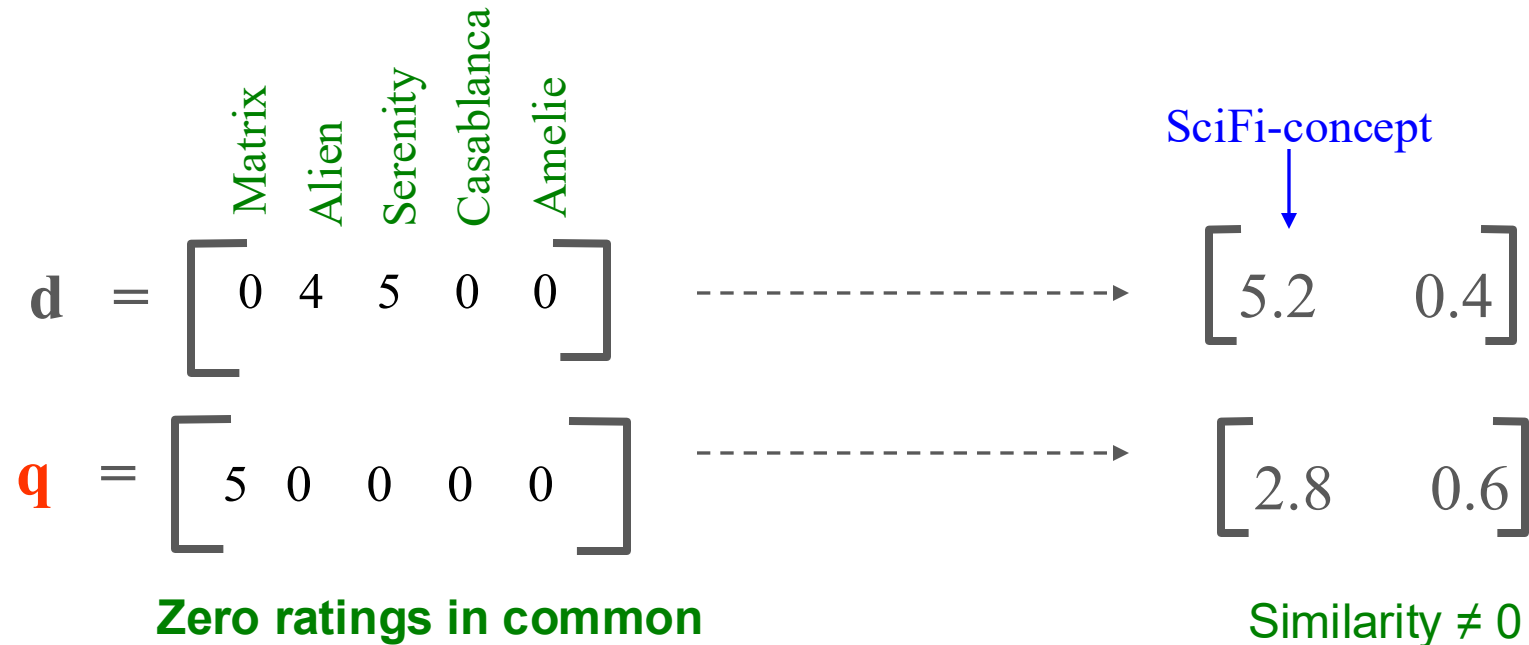
E.g.:

$$\mathbf{d} = \begin{matrix} & \begin{matrix} \text{Matrix} \\ \text{Alien} \\ \text{Serenity} \\ \text{Casablanca} \\ \text{Amelie} \end{matrix} \\ \begin{matrix} 0 & 4 & 5 & 0 & 0 \end{matrix} \end{matrix} \times \begin{matrix} \begin{bmatrix} 0.56 & 0.12 \\ 0.59 & -0.02 \\ 0.56 & 0.12 \\ 0.09 & -0.69 \\ 0.09 & -0.69 \end{bmatrix} \\ \text{movie-to-concept} \\ \text{similarities (V)} \end{matrix} = \begin{matrix} \text{SciFi-concept} \\ \downarrow \\ \begin{bmatrix} 5.2 & 0.4 \end{bmatrix} \end{matrix}$$

Case study: How to query?

62

- **Observation:** User d that rated (*Alien*, *Serenity*) will be **similar** to user q that rated (*Matrix*), although d and q have **zero ratings in common**!



Exercise

63

- Suppose Leslie assigns rating 5 to Alien and rating 2 to Titanic, giving us a representation of Leslie in “movie space” of $[0, 5, 0, 0, 2]$. Find the representation of Leslie in concept space. What does that representation predict about how well Leslie would like the other movies appearing in our example data?

$$\begin{array}{c}
 \begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 3 & 3 & 3 & 0 & 0 \\ 4 & 4 & 4 & 0 & 0 \\ 5 & 5 & 5 & 0 & 0 \\ 0 & 0 & 0 & 4 & 4 \\ 0 & 0 & 0 & 5 & 5 \\ 0 & 0 & 0 & 2 & 2 \end{bmatrix} = \begin{bmatrix} .14 & 0 \\ .42 & 0 \\ .56 & 0 \\ .70 & 0 \\ 0 & .60 \\ 0 & .75 \\ 0 & .30 \end{bmatrix} \begin{bmatrix} 12.4 & 0 \\ 0 & 9.5 \end{bmatrix} \begin{bmatrix} .58 & .58 & .58 & 0 & 0 \\ 0 & 0 & 0 & .71 & .71 \end{bmatrix} \\
 M \qquad \qquad \qquad U \qquad \qquad \qquad \Sigma \qquad \qquad \qquad V^T
 \end{array}$$

SVD: Drawbacks

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- + **Optimal low-rank approximation**
in terms of Frobenius norm
- **Interpretability problem:**
 - A singular vector specifies a linear combination of all input columns or rows
- **Lack of sparsity:**
 - Singular vectors are **dense!**

The diagram illustrates the SVD decomposition of a matrix A into three components: U , Σ , and V^T . Matrix A is represented by a rectangle containing 6 black dots, indicating it is sparse. It is followed by an equals sign. To the right of the equals sign is matrix U , a tall rectangle filled with many black dots, indicating it is dense. Next is matrix Σ , a small square containing a single black dot, representing the singular values. Finally, matrix V^T is shown as a horizontal rectangle filled with many black dots, also indicating it is dense.

CUR Decomposition



Frobenius norm: $\|X\|_F = \sqrt{\sum_{ij} X_{ij}^2}$

CUR Decomposition

66

- **Goal: Express A as a product of matrices C,U,R**

Make $\|A - C \cdot U \cdot R\|_F$ small

- **“Constraints” on C and R:**

$$\left(\begin{array}{c|c|c} \text{red} & \text{yellow} & \text{gray} \\ \hline & A & \end{array} \right) \approx \left(\begin{array}{c|c|c|c|c|c|c} \text{red} & \text{red} & \text{red} & \text{yellow} & \text{gray} & \text{gray} & \text{gray} \\ \hline & C & \end{array} \right) \cdot \left(\begin{array}{c} U \end{array} \right) \cdot \left(\begin{array}{c} R \end{array} \right)$$

A C U R

Frobenius norm: $\|X\|_F = \sqrt{\sum_{ij} X_{ij}^2}$

CUR Decomposition

67

- **Goal: Express A as a product of matrices C,U,R**

Make $\|A - C \cdot U \cdot R\|_F$ small

- **“Constraints” on C and R:**

$$\begin{pmatrix} \text{red bar} \\ \text{gray bar } A \\ \text{yellow bar} \end{pmatrix} \approx \begin{pmatrix} C \end{pmatrix} \cdot \begin{pmatrix} U \end{pmatrix} \cdot \begin{pmatrix} \text{red bar} \\ \text{red bar} \\ \text{red bar} \\ \text{gray bar } R \\ \text{gray bar} \\ \text{yellow bar} \end{pmatrix}$$

A
C
U
R


Pseudo-inverse of the intersection of C and R

CUR: How it Works

68

- **Sampling columns (similarly for rows):**

Input: matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$, sample size c

Output: $\mathbf{C}_d \in \mathbb{R}^{m \times c}$

1	1	1	0	0
3	3	3	0	0
4	4	4	0	0
5	5	5	0	0
0	2	0	4	4
0	0	0	5	5
0	1	0	2	2

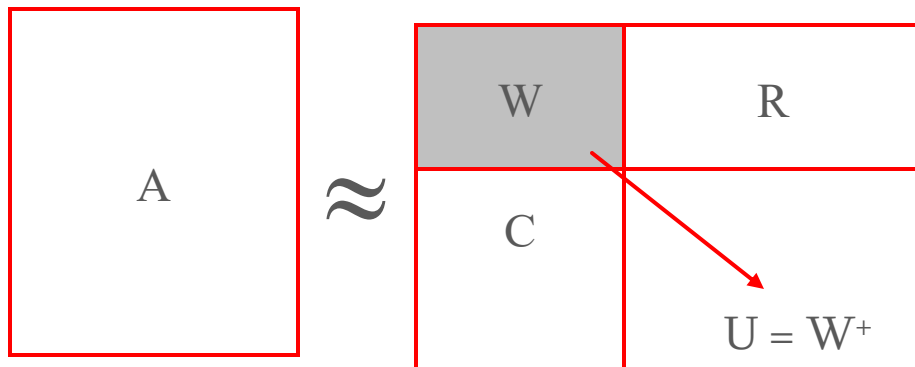
1. for $x = 1 : n$ [column distribution]
2. $P(x) = \sum_i \mathbf{A}(i, x)^2 / \sum_{i,j} \mathbf{A}(i, j)^2$
3. for $i = 1 : c$ [sample columns]
4. Pick $j \in 1 : n$ based on distribution $P(x)$
5. Compute $\mathbf{C}_d(:, i) = \mathbf{A}(:, j) / \sqrt{cP(j)}$

Note this is a randomized algorithm, same column can be sampled more than once

Computing U

69

- Let $\mathbf{W}$ be the “intersection” of sampled columns $\mathbf{C}$ and rows $\mathbf{R}$
 - Let SVD of $\mathbf{W} = \mathbf{X} \mathbf{Z} \mathbf{Y}^T$
- **Then:** $\mathbf{U} = \mathbf{W}^+ = \mathbf{Y} \mathbf{Z}^+ \mathbf{X}^T$
 - $\mathbf{Z}^+$: **reciprocals of non-zero singular values:** $Z_{ii}^+ = 1/Z_{ii}$
 - $\mathbf{W}^+$ is the “**pseudoinverse**”



Why pseudoinverse works?

$\mathbf{W} = \mathbf{X} \mathbf{Z} \mathbf{Y}$ then $\mathbf{W}^{-1} = \mathbf{X}^{-1} \mathbf{Z}^{-1} \mathbf{Y}^{-1}$

Due to orthonormality

$\mathbf{X}^{-1} = \mathbf{X}^T$ and $\mathbf{Y}^{-1} = \mathbf{Y}^T$

Since $\mathbf{Z}$ is diagonal $\mathbf{Z}^{-1} = 1/Z_{ii}$

Thus, if $\mathbf{W}$ is nonsingular,
pseudoinverse is the true
inverse

CUR: Provably good approx. to SVD

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- **For example:**

- Select $c = O\left(\frac{k \log k}{\epsilon^2}\right)$ columns of A using **ColumnSelect** algorithm
- Select $r = O\left(\frac{k \log k}{\epsilon^2}\right)$ rows of A using **ColumnSelect** algorithm
- Set $U = W^+$

- **Then:**

with probability 98%

$$\|A - \overset{\text{CUR error}}{CUR}\|_F \leq (2 + \epsilon) \|A - \overset{\text{SVD error}}{A_k}\|_F$$

In practice:

Pick $4k$ cols/rows

for a “rank- k ” approximation

CUR: Pros & Cons

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+ Easy interpretation

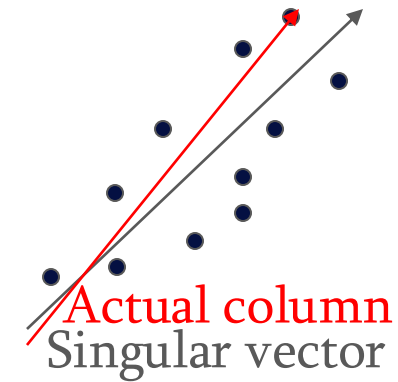
- Since the basis vectors are actual columns and rows

+ Sparse basis

- Since the basis vectors are actual columns and rows

- Duplicate columns and rows

- Columns of large norms will be sampled many times

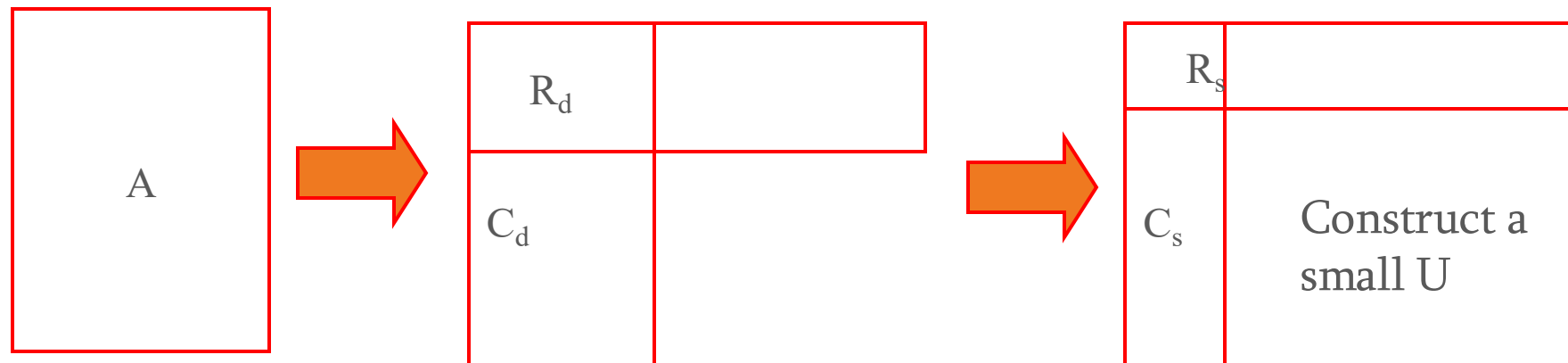


Solution

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- **If we want to get rid of the duplicates:**

- Throw them away
- Scale (multiply) the columns/rows by the square root of the number of duplicates



SVD vs. CUR

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SVD: $A = U \Sigma V^T$

Annotations for SVD:

- A : Huge but sparse
- U : Big and dense
- Σ : sparse and small
- V^T : Big and dense

CUR: $A = C U R$

Annotations for CUR:

- A : Huge but sparse
- C : Big but sparse
- U : dense but small
- R : Big but sparse

SVD vs. CUR: Simple Experiment

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- **DBLP bibliographic data**

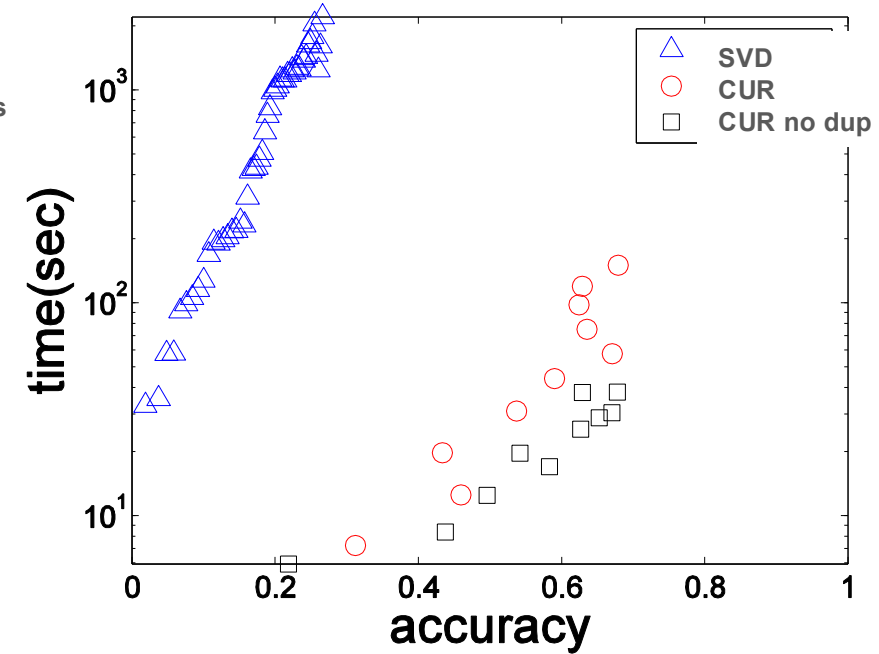
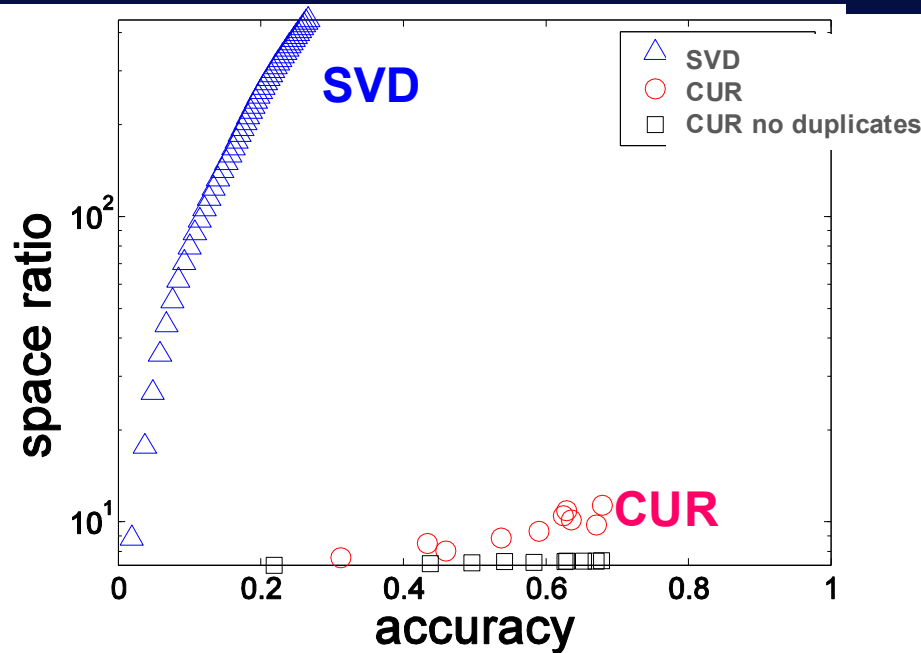
- Author-to-conference big sparse matrix
- A_{ij} : Number of papers published by author i at conference j
- 428K authors (rows), 3659 conferences (columns)
 - Very sparse

- **Want to reduce dimensionality**

- How much time does it take?
- What is the reconstruction error?
- How much space do we need?

Results: DBLP- big sparse matrix

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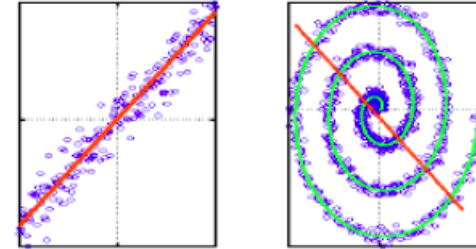
- **Accuracy.**
 - $1 - \text{relative sum squared errors}$
- **Space ratio:**
 - $\text{\#output matrix entries} / \text{\#input matrix entries}$
- **CPU time**

What about linearity assumption?

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- **SVD is limited to linear projections:**

- Lower-dimensional linear projection that preserves Euclidean distances

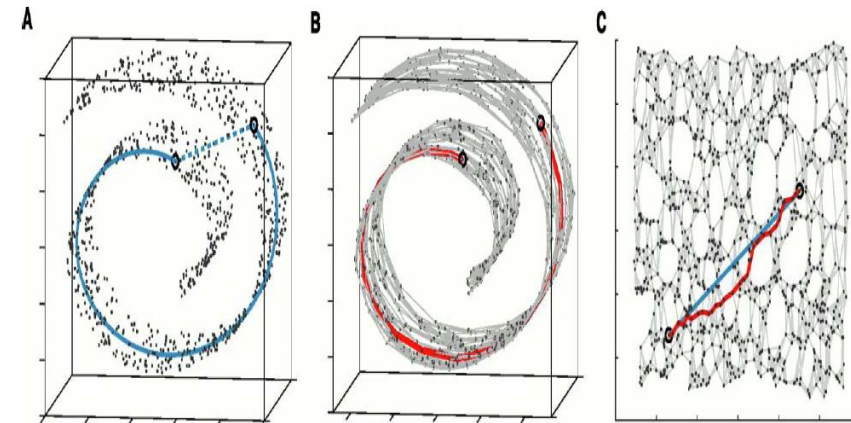


- **Non-linear methods: Isomap**

- Data lies on a nonlinear low-dim curve aka manifold
 - Use the distance as measured along the manifold

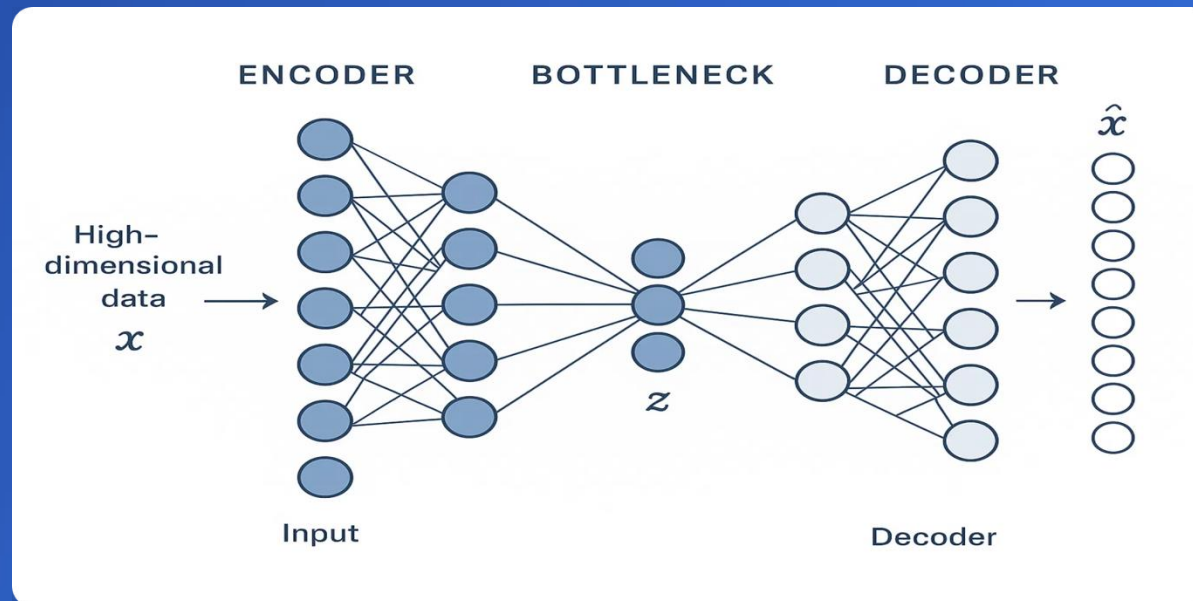
- **How?**

- Build adjacency graph
- Geodesic distance is graph distance
- SVD/PCA the graph pairwise distance matrix



Autoencoders and Sequence Methods for Dimensionality Reduction

Neural Network Approaches to Data Compression and Feature Learning



What is an Autoencoder?

Definition

An **autoencoder** is a type of artificial neural network used to learn efficient data representations in an unsupervised manner.

The network is trained to copy its input to its output, forcing it to learn a compressed representation of the data in a hidden layer.

Key Characteristics:

Unsupervised learning - No labeled data required

Self-supervised - Input serves as the target output

Dimensionality reduction

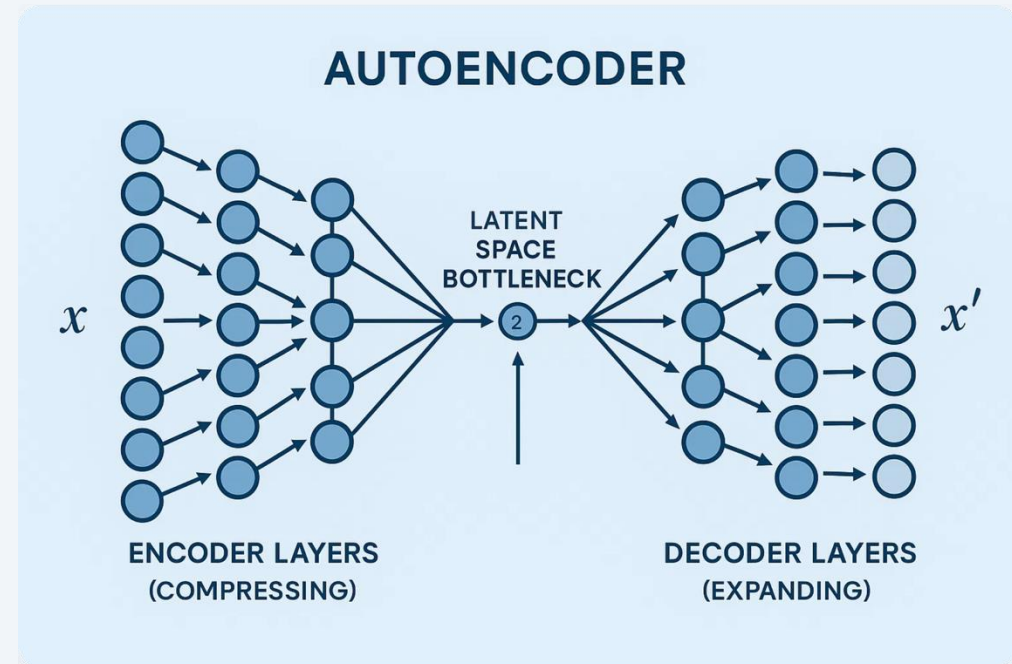
- Compresses data into lower dimensions

Feature learning

- Automatically discovers important patterns

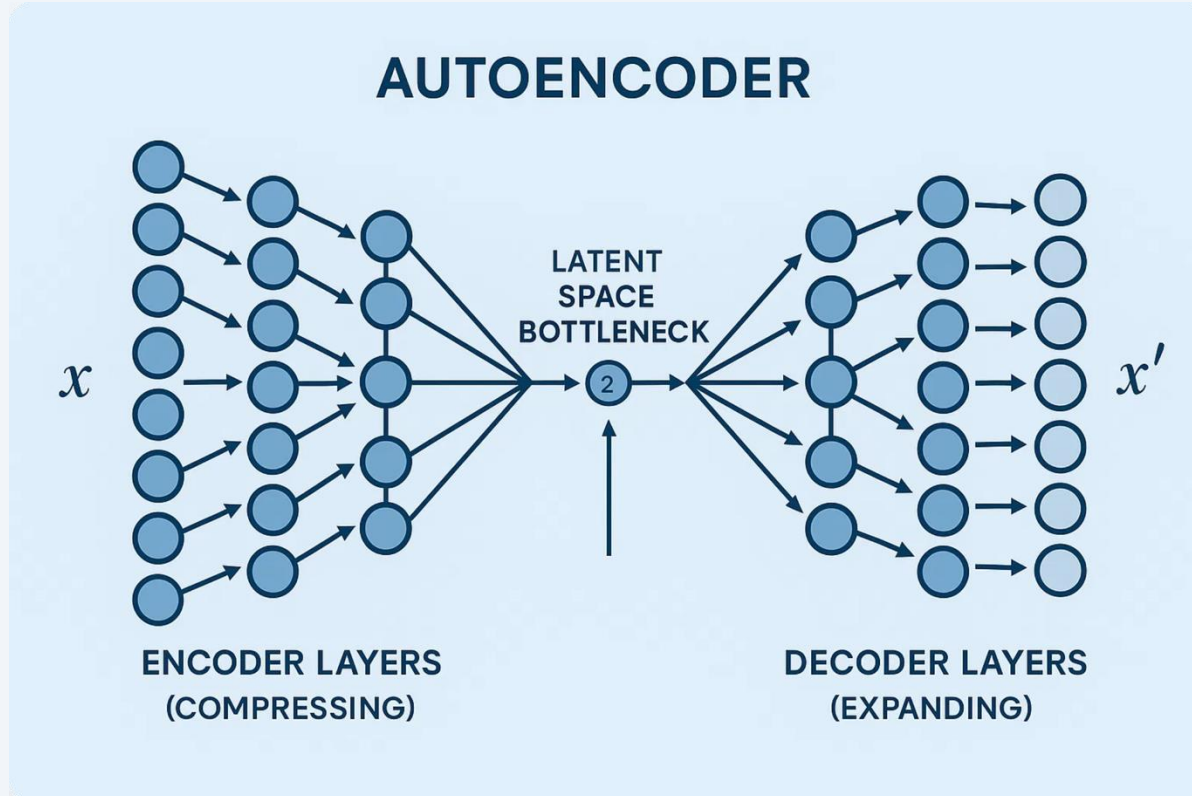
Reconstruction

- Attempts to recreate original input



Basic autoencoder architecture showing encoder, latent space, and decoder components

Autoencoder Architecture



Mathematical Formulation

Encoder: $z = f(x) = \sigma(Wx + b)$
Decoder: $x' = g(z) = \sigma(W'z + b')$

Key Components

✂ Encoder

Compresses input data x into lower-dimensional latent representation z .

Latent Space

Compressed representation containing the most important features.

Decoder

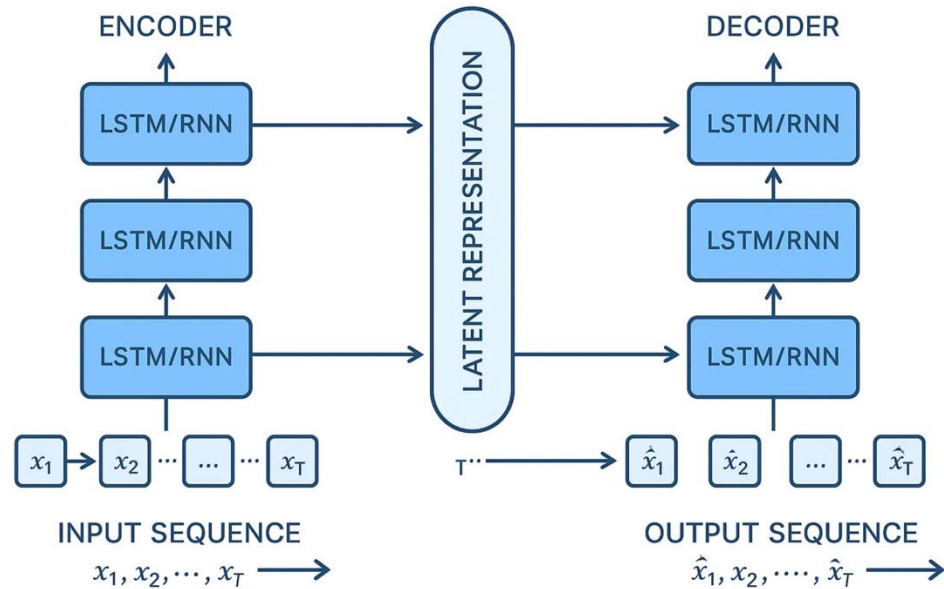
Reconstructs original data from latent representation.

Loss Function

Measures difference between input x and reconstructed output x' .

Sequence Autoencoders

SEQUENCE-TO-SEQUENCE AUTOENCODER



*Sequence-to-sequence autoencoder architecture with
LSTM/RNN cells*

Sequence-to-Sequence Autoencoders

Specialized autoencoders designed to process sequential data like time series, text, or audio by capturing temporal dependencies.

≡ Recurrent Cells:

Uses LSTM or GRU cells to maintain memory of previous timesteps

✂ Encoder:

Processes input sequence to create fixed-length latent vector

Decoder:

Reconstructs original sequence from latent representation

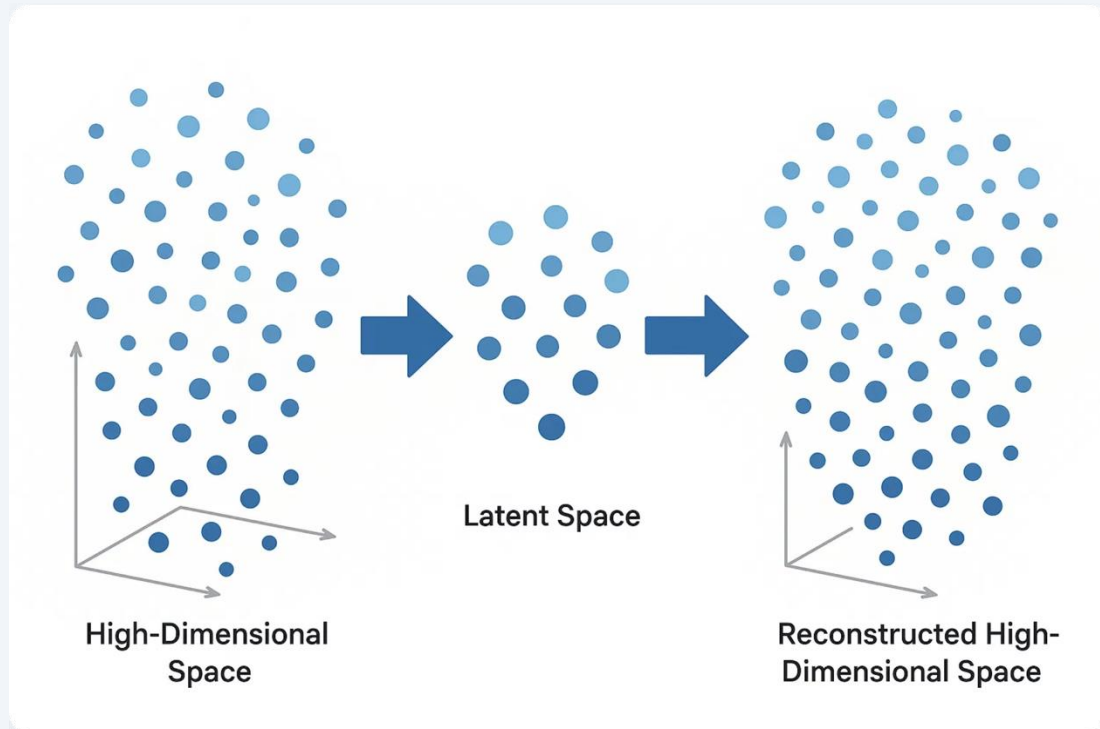
Attention Mechanism:

Optional component to focus on relevant parts of input sequence

Applications:

Time series forecasting, anomaly detection, speech processing, and text generation

Dimensionality Reduction Process



Transformation from high-dimensional space to low-dimensional latent space and back to reconstructed space

How Autoencoders Reduce Dimensions

1 Data Encoding

High-dimensional input data ($x \in \mathbb{R}^n$) is compressed through multiple neural network layers, progressively reducing dimensions.

2 Latent Representation

Data is mapped to a lower-dimensional latent space ($z \in \mathbb{R}^m$, where $m \ll n$), capturing the most important features.

3 Reconstruction

The decoder attempts to reconstruct the original data from the compressed representation, minimizing information loss.

Benefits of Autoencoder Dimensionality Reduction

Non-linear transformations Feature learning
Noise filtering Adaptable architecture

Applications and Use Cases

Image Processing

Image denoising, compression, colorization, and super-resolution. Convolutional autoencoders excel at preserving spatial relationships.

Time Series Analysis

Forecasting, anomaly detection in sensor data, and extracting temporal patterns using sequence autoencoders with LSTM/GRU cells.

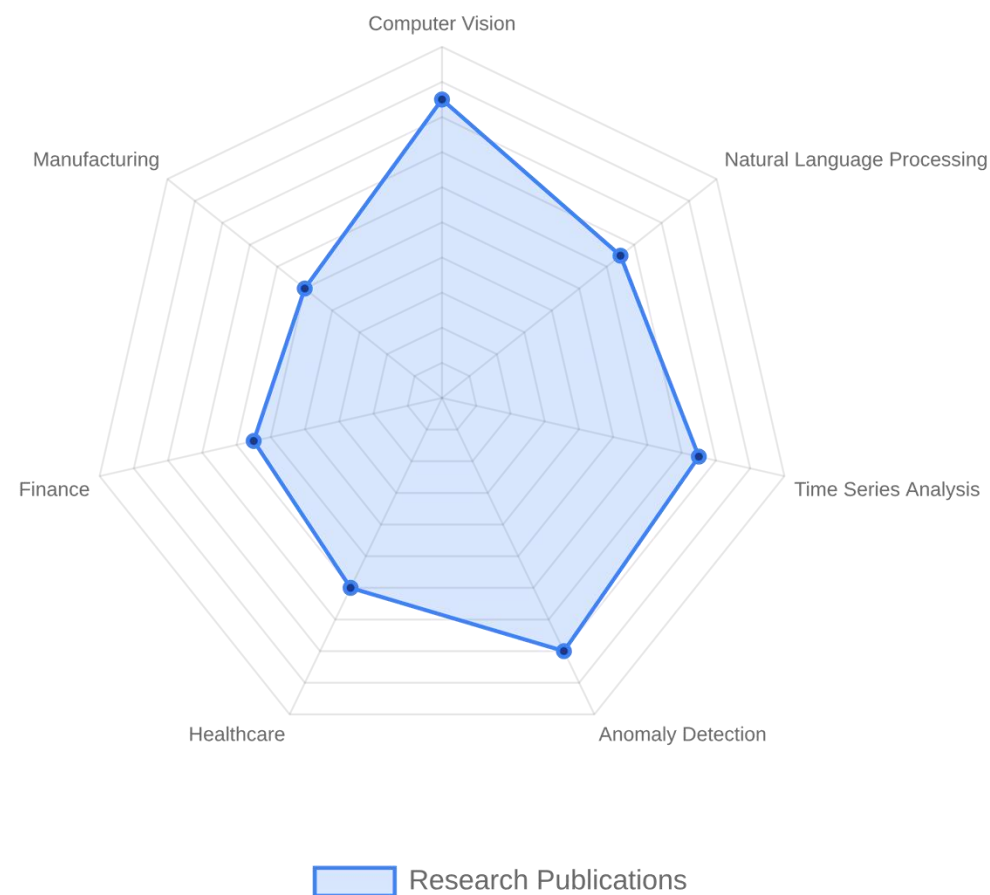
Anomaly Detection

Identifying outliers by measuring reconstruction error. Higher error -> anomalous data points in fraud detection

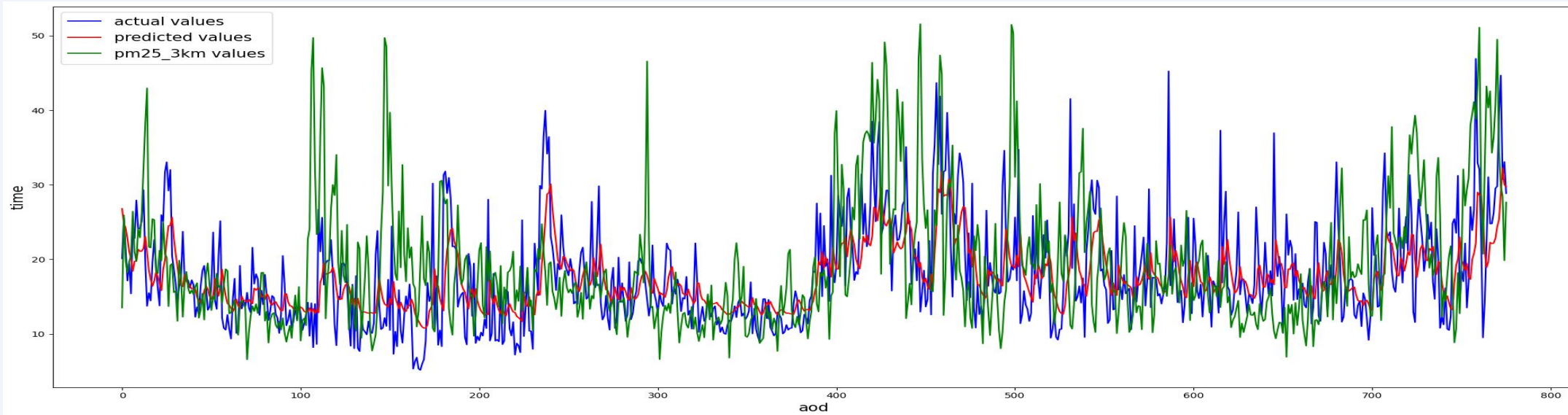
Feature Learning

Extracting meaningful features from raw data for downstream tasks like classification,

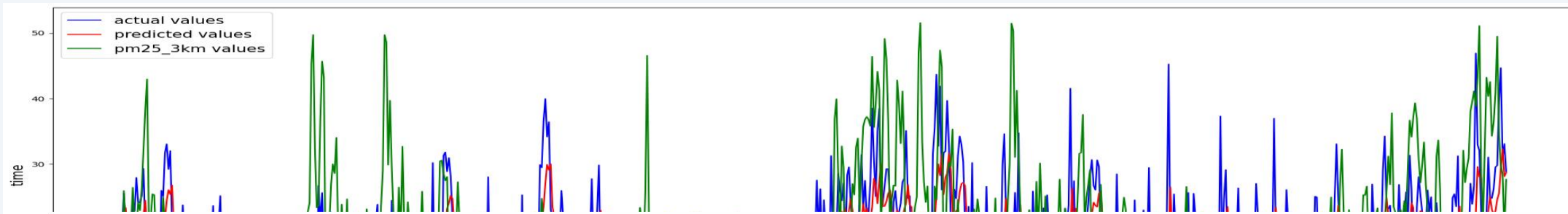
Autoencoder Applications by Domain



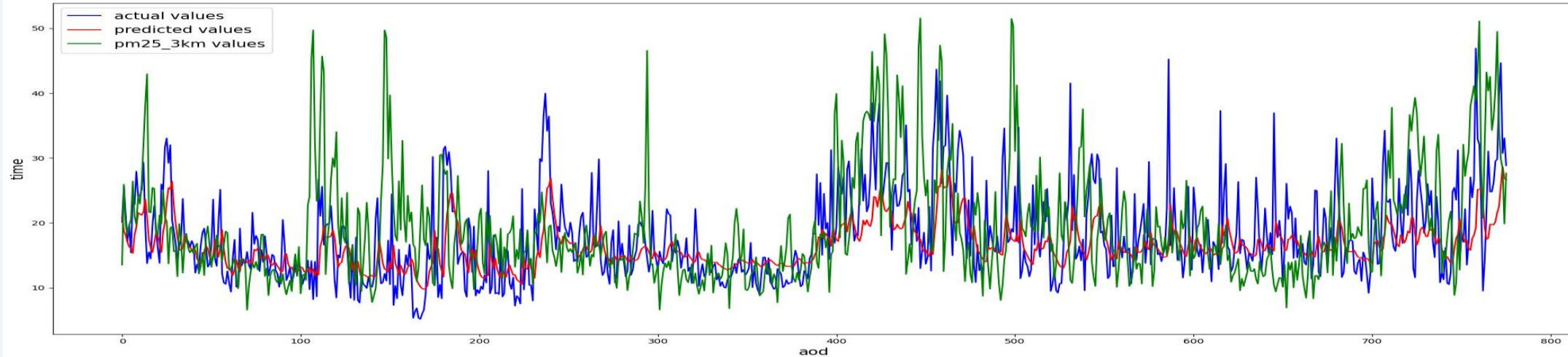
LSTMSeq2Seq
+ LSTM
MAE: 5.81



GRUSeq2Seq +
LSTM
MAE: 5.41



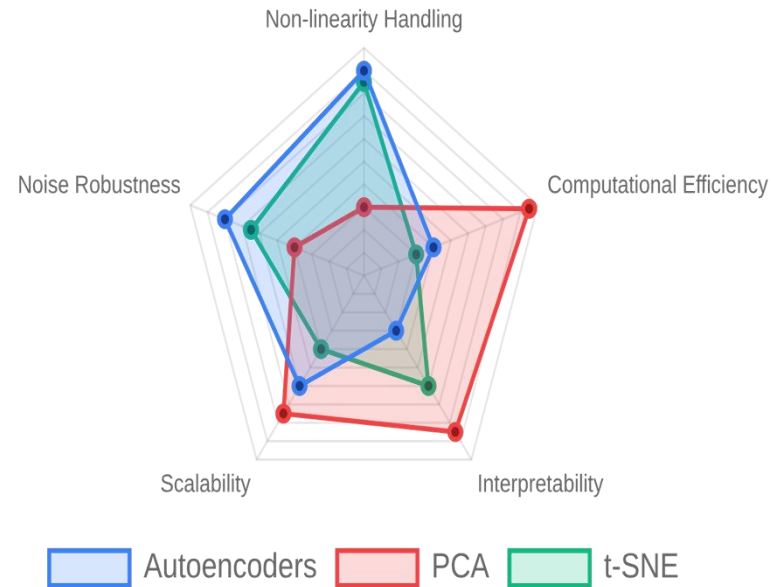
CNN-
LSTMSeq2Seq
+ LSTM
MAE: 5.47



Advantages and Limitations

Advantages

Limitations



patterns remain intact.

● Noise Reduction

Denoising autoencoders can filter out noise and corruption in data, improving quality of representations.

each dimension represents.

● Data Requirements

Requires substantial amounts of training data to learn effective representations and avoid overfitting.

Comparison with Other Dimensionality Reduction Methods

Basic operations on matrices



Types of matrices

Row matrix or vector

Any number of columns but only one row

$$\begin{bmatrix} 1 & 1 & 6 \end{bmatrix} \quad \begin{bmatrix} 0 & 3 & 5 & 2 \end{bmatrix}$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \cdots a_{1n} \end{bmatrix}$$

Column matrix or vector

Number of rows may be any integer but the number of columns is always 1

$$\begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix} \quad \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix}$$

Types of matrices

Diagonal matrix

A square matrix where all the elements are zero except those on the main diagonal

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Square matrix

Number of rows = number of columns

$$\begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 9 \end{bmatrix}$$

Types of matrices

Unit or Identity matrix - I

A diagonal matrix with ones on the main diagonal

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} a_{ij} & 0 \\ 0 & a_{ij} \end{bmatrix}$$

Null (zero) matrix - 0

All elements in the matrix are zero

$$\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Types of matrices

Triangular matrix

A square matrix whose elements above or below the main diagonal are all zero

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 5 & 2 & 3 \end{bmatrix} \quad \begin{bmatrix} 1 & 8 & 9 \\ 0 & 1 & 6 \\ 0 & 0 & 3 \end{bmatrix}$$

Upper/Lower triangular matrix

A square matrix whose elements below/above the main diagonal are all zero

Types of matrices

Scalar matrix

A diagonal matrix whose main diagonal elements are equal to the same scalar

A scalar is defined as a single number or constant

A square matrix whose elements above or below the main diagonal are all zero

$$\begin{bmatrix} a_{ij} & 0 & 0 \\ 0 & a_{ij} & 0 \\ 0 & 0 & a_{ij} \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 6 & 0 & 0 & 0 \\ 0 & 6 & 0 & 0 \\ 0 & 0 & 6 & 0 \\ 0 & 0 & 0 & 6 \end{bmatrix}$$

Addition and Subtraction of matrices

- Must be of the same order, mxn:

$$A + B = \begin{bmatrix} a_{ij} + b_{ij} \end{bmatrix}$$

$$A - B = \begin{bmatrix} a_{ij} - b_{ij} \end{bmatrix}$$

$$\begin{bmatrix} 4 & -3 & 1 \\ 0 & 5 & -2 \\ 5 & -6 & 0 \end{bmatrix} + \begin{bmatrix} -1 & 2 & 3 \\ 6 & -7 & 9 \\ 0 & -4 & 8 \end{bmatrix}$$

Addition

$$\begin{bmatrix} 5 & 5 & 4 \\ 6 & 12 & 11 \\ 5 & 10 & 8 \end{bmatrix}$$

Substraction

$$\begin{bmatrix} 3 & 1 & -2 \\ -6 & -2 & -7 \\ 5 & 2 & -8 \end{bmatrix}$$

Scalar Multiplication

- The **scalar product** of a number k and a matrix A
- k : **scalar**.

$$kA = [ka_{ij}]$$

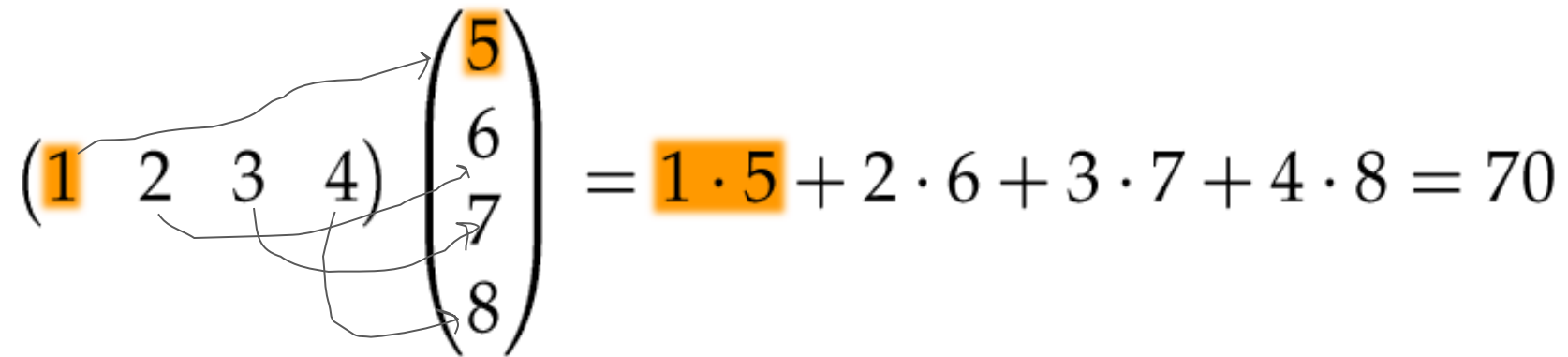
Transpose of a matrix

$$A = {}_2A^3 = \begin{bmatrix} 2 & 4 & 7 \\ 5 & 3 & 1 \end{bmatrix}$$

$$A^T = {}_2A^{3T} = \begin{bmatrix} 2 & 5 \\ 4 & 3 \\ 7 & 1 \end{bmatrix}$$

Product of a Row Matrix and a Column Matrix

- Number of columns of a row matrix = the number of rows of a column matrix
- $R(1 \times 4) \times C(4 \times 1) = M(1 \times 1)$


$$(1 \ 2 \ 3 \ 4) \begin{pmatrix} 5 \\ 6 \\ 7 \\ 8 \end{pmatrix} = 1 \cdot 5 + 2 \cdot 6 + 3 \cdot 7 + 4 \cdot 8 = 70$$

Revenue of a car dealer

- A car dealer sells four model types: A,B,C,D. On a given week, this dealer sold 10 cars of model A, 5 of model B, 8 of model C and 3 of model D. The selling prices of each automobile are respectively \$12,500, \$11,800, \$15,900 and \$25,300. Represent the data using matrices and use matrix multiplication to find the total revenue.

$$\begin{bmatrix} 10 & 5 & 8 & 3 \end{bmatrix} \begin{bmatrix} 12,500 \\ 11,800 \\ 15,900 \\ 25,300 \end{bmatrix} = [10(12,500) + 5(11,800) + 8(15,900) + 3(25,300)] = [387,100]$$

Matrix Product

- **A** is an $m \times p$ matrix
- **B** is a $p \times n$ matrix,
- **Matrix product** of A and B denoted by AB is an $m \times n$ matrix

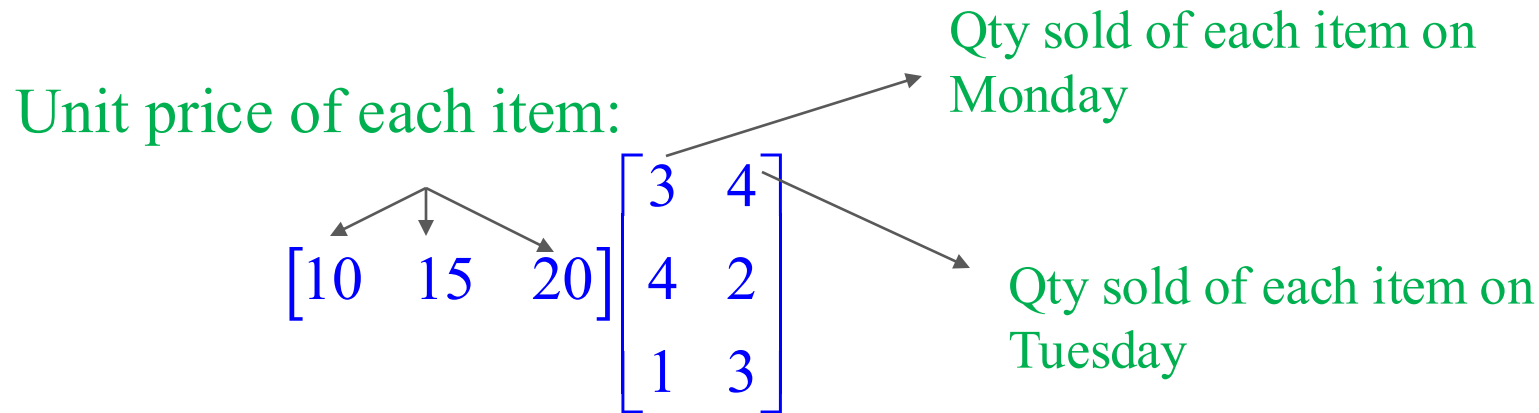
Is Matrix Multiplication Commutative?
 $A(3 \times 4)B(4 \times 3) = BA$?

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \\ 10 & 11 & 12 \end{pmatrix} = \begin{pmatrix} 70 & 80 & 90 \\ 158 & 184 & 210 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \\ 10 & 11 & 12 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{pmatrix} \text{ is not defined}$$

Practical application

- Suppose you a business owner and sell clothing. The following represents the number of items sold and the cost for each item: Use matrix operations to determine the total revenue over the two days:
- Monday: 3 T-shirts at \$10 each, 4 hats at \$15 each, and 1 pair of shorts at \$20.
Tuesday: 4 T-shirts at \$10 each, 2 hats at \$15 each, and 3 pairs of shorts at \$20.



Orthogonal matrices (orthonormal)

- Each column vector is orthogonal to every other column vector in the matrix.
- Each column in the matrix has length 1
- $M^T M = I$
- Columns of M form an orthonormal basis of $\mathbb{R}^n$

$$\begin{bmatrix} -0.8 & 0.6 \\ 0.6 & 0.8 \end{bmatrix}$$

Product AB of two orthogonal $n \times n$ matrices is orthogonal

Example

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$$M = \begin{bmatrix} 1 & 3 \\ 3 & 5 \\ 5 & 3 \\ 0 & 4 \\ 1 & 5 \end{bmatrix}$$

1. Calculate the eigenvectors and eigenvalues $M^T M$.
2. Reduce M to one dimension